CHAPTER

11

Nanometer Design Studies

The previous chapters of this book have taken us on a "scenic" route through the world of analog circuits, presenting important concepts and useful topologies. We have occasionally made *design* efforts, but only on a small scale. In this chapter, we embark upon two comprehensive designs so as to appreciate the mindset that an analog designer must uphold and the multitude of tasks that he or she must complete for a given circuit. The designs are carried out in 40-nm CMOS technology with a 1-V supply. The reader is encouraged to review the op amp design examples in Chapter 9 before starting this chapter.

We begin with a brief look at the imperfections of nanometer devices and the design procedures to achieve certain transistor parameters. We then delve into the design of an op amp and, through simulations, optimize its performance. Finally, we deal with the design of a high-speed, high-precision amplifier and pursue various techniques to achieve a low power dissipation.

11.1 ■ Transistor Design Considerations

In Chapter 2, we studied the basic operation of MOSFETs and included a few second-order effects. Our investigation has produced a large-signal model (consisting of the triode-region quadratic equation and the saturation-region square-law relation), which becomes necessary in two cases: (1) when the transistor experiences large voltage (or current) changes due to the input or output signals, disobeying the small-signal model, or (2) when the transistor must be biased, requiring certain terminal voltages so as to carry a specified current. In analog design, the former case occurs occasionally, while the latter almost always.

The large-signal behavior of nanometer MOSFETs significantly departs from the "long-channel" model that we have developed. As a result of technology scaling, i.e., the shrinkage of MOS dimensions, several effects besides those studied in Chapter 2 manifest themselves, thereby altering the I/V characteristics. As an example, Fig. 11.1 plots the actual I_D-V_{DS} characteristics of an NFET with $W/L = 5\ \mu$m/40 nm and $V_{TH} \approx 300$ mV (using a BSIM4 model) against a "best-fit" long-channel square-law approximation. We observe that the two diverge considerably. Thus, even if we are not interested in the large-signal analysis of a circuit, we still face the problem of bias calculations using the square-law model.

In this section, we briefly consider a few "short-channel" effects that make the long-channel model inaccurate. A detailed treatment of short-channel effects is deferred to Chapter 17. It is important to note that the MOS small-signal model developed in Chapter 2 still holds for short-channel devices and, as seen throughout this book, suffices for the initial analysis of many analog circuit blocks. However, the expressions relating g_m and r_O to the bias conditions must be revised.

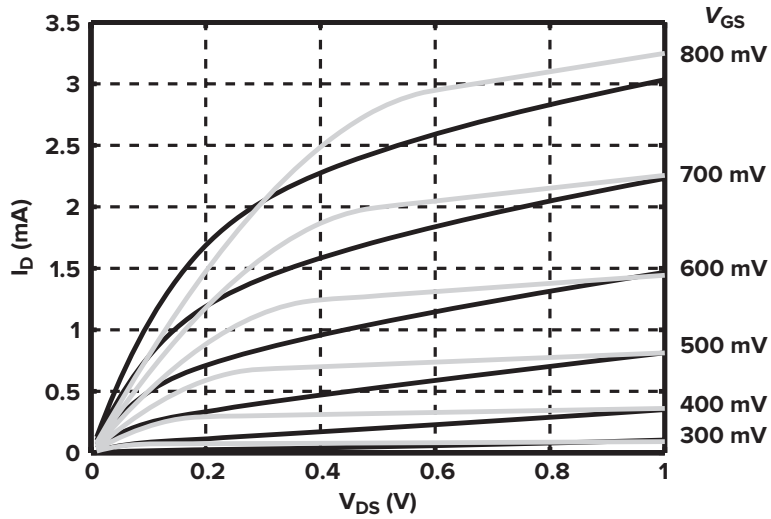


Figure 11.1 I-V characteristics of an actual 5-μm/40-nm device (black curves) and a best-fit square-law device (gray curves). (V_{GS} is incremented from 300 mV to 800 mV in 100-mV steps.)

The characteristics shown in Fig. 11.1 exhibit severe channel-length modulation for the actual 40-nm devices, making it difficult to distinguish between the triode and saturation regions. But we can associate a "knee" point with each curve as a rough boundary. Figure 11.2 plots the actual 40-nm device characteristics for a narrower V_{GS} range, namely, $V_{GS} - V_{TH} = 50$ mV, 100 mV, $\cdots$, 350 mV. We observe knee points below $V_{DS} = 0.2$ V. (Here, $W = 5$ μm and $V_{TH} \approx 200$ mV.)

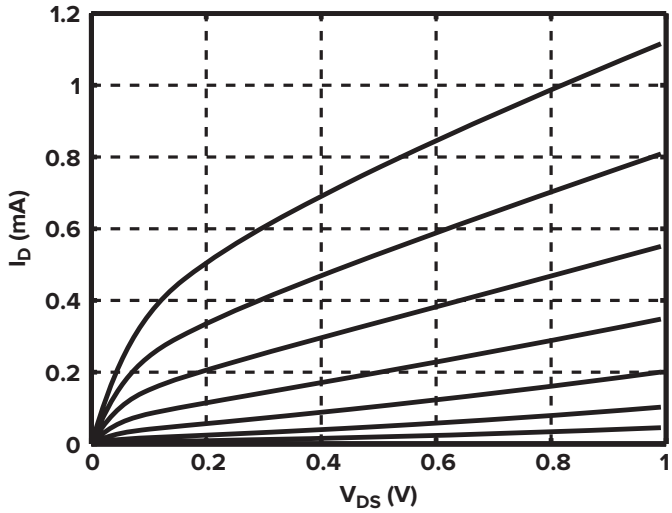


Figure 11.2 I-V characteristics of a 5-μm/40-nm device for $V_{GS} - V_{TH} = 50, \cdots, 350$ mV.

11.2 ■ Deep-Submicron Effects

Among various short-channel effects, two are particularly important at this stage of our studies; both relate to the mobility of the carriers in the channel. Recall that we have assumed that the carrier velocity is given by $v = \mu E$, where E denotes the electric field. We revisit this assumption here.

Velocity Saturation In a MOSFET, as V_{DS} and hence the electric field along the source-drain path increase, v does not rise proportionally (Fig. 11.3).

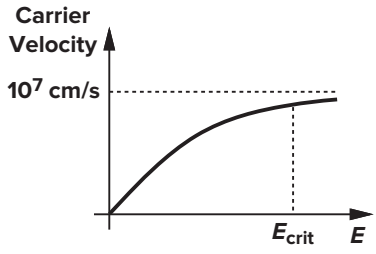


Figure 11.3 Velocity saturation at high electric fields.

We say that the carriers experience "velocity saturation" or, equivalently, that the mobility (the slope of v versus E) *falls*. This effect has arisen because the length of MOSFETs has shrunk from, say, 1 μm to 40 nm (a factor of 25) while the allowable drain-source voltage has decreased from 5 V to about 1 V. The lateral electric field has thus exceeded E_{crit} ($\approx$ 1 V/μm) in Fig. 11.3.

We deal with the modeling of velocity saturation in Chapter 17, but let us consider an extreme case here: suppose the charge carriers reach the saturated velocity, v_{sat}, as soon as they depart from the source. Since $I = Q_d \cdot v$, where Q_d is the charge density (per unit length) and given by $WC_{ox}(V_{GS} - V_{TH})$, we have

$$I_D = WC_{ox}(V_{GS} - V_{TH})v_{sat} \tag{11.1}$$

Extreme velocity saturation therefore creates three departures from the square-law behavior. First, the device carries a current that is *linearly* proportional to the overdrive and independent of the channel length.[1] Second, I_D reaches saturation even for $V_{DS} < V_{GS} - V_{TH}$ (Fig. 11.4). As evident in Fig. 11.2, the knee points occur at relatively small V_{DS}'s even as the overdrive reaches 350 mV. Third, the transconductance of a fully velocity-saturated MOSFET emerges as

$$g_m = \frac{\partial I_D}{\partial V_{GS}}|_{V_{DSconst}} \tag{11.2}$$

$$= WC_{ox}v_{sat} \tag{11.3}$$

a relatively *constant* value versus I_D or V_{GS}. For example, in the plots of Fig. 11.2, the change in I_D is fairly constant as the overdrive increments from 250 mV to 300 mV and from 300 mV to 350 mV.

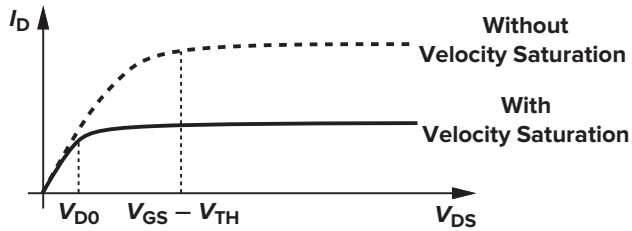


Figure 11.4 Premature saturation of drain current due to velocity saturation.

Mobility Degradation with Vertical Field The mobility of the charge carriers in the channel also declines as the gate-source voltage and the *vertical* field increase (Fig. 11.5).

[1] So long as L is small enough and V_{DS} large enough to cause velocity saturation.

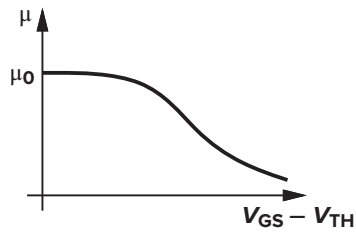


Figure 11.5 Reduction of mobility due to vertical electric field.

What is the impact of this mobility degradation on the device transconductance? We intuitively expect that g_m no longer follows the linear relationship, $g_m = \mu C_{ox}(W/L)(V_{GS} - V_{TH})$, with the overdrive voltage. Figure 11.6 displays this behavior for the 5-μm/40-nm NFET mentioned above.

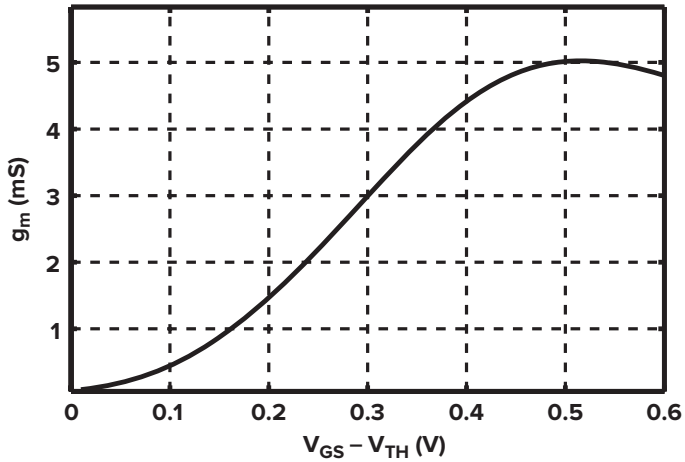


Figure 11.6 Transconductance as a function of overdrive voltage.

▶ **Example 11.1**

We approximate the mobility plot of Fig. 11.5 by

$$\mu = \frac{\mu_0}{1 + \theta(V_{GS} - V_{TH})} \tag{11.4}$$

where θ is a proportionality factor with a dimension of (voltage)$^{-1}$. Determine the transconductance of a MOSFET that suffers from this type of mobility degradation.

Solution

We write

$$I_D = \frac{1}{2}\frac{\mu_0 C_{ox}}{1 + \theta(V_{GS} - V_{TH})}\frac{W}{L}(V_{GS} - V_{TH})^2 \tag{11.5}$$

and hence

$$g_m = \mu_0 C_{ox}\frac{W}{L}\frac{(\theta/2)(V_{GS} - V_{TH})^2 + V_{GS} - V_{TH}}{[1 + \theta(V_{GS} - V_{TH})]^2} \tag{11.6}$$

As expected, for $\theta(V_{GS} - V_{TH}) \ll 1$, we have $g_m \approx \mu_0 C_{ox}(W/L)(V_{GS} - V_{TH})$. At the other extreme, if $(V_{GS} - V_{TH}) \gg 2/\theta$, then g_m approaches a constant value: $g_m \approx (1/2)\mu_0 C_{ox}(W/L)/\theta$.

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In the general case, the degradation of the mobility due to both lateral and vertical fields (V_{DS} and V_{GS}, respectively) must be considered. Nonetheless, the simple results derived above suffice for most of our studies in analog design.

11.3 ■ Transconductance Scaling

Device transconductances manifest themselves in almost every analog circuit. Suppose a transistor operates in the saturation region but does not provide the required transconductance. The g_m equations in Chapter 2,

$$g_m = \mu_n C_{ox} \frac{W}{L}(V_{GS} - V_{TH}) \tag{11.7}$$

$$= \sqrt{2\mu_n C_{ox} \frac{W}{L} I_D} \tag{11.8}$$

$$= \frac{2I_D}{V_{GS} - V_{TH}} \tag{11.9}$$

suggest that adjustments in three parameters, namely, W/L, $V_{GS}-V_{TH}$, or I_D, can scale g_m. We study these scenarios, assuming for now a long-channel device and hence $I_D \approx (1/2)\mu_n C_{ox}(W/L)(V_{GS} - V_{TH})^2$. That is, $V_{GS} - V_{TH} \approx \sqrt{2I_D/(\mu_n C_{ox} W/L)}$. In each case, we keep one of the parameters constant and vary the other two.

From (11.7), we can increase W/L while keeping $V_{GS} - V_{TH}$ constant. In this case, both g_m and I_D linearly scale with W/L (why?) [Fig. 11.7(a)], and so does the power consumption. Alternatively, we

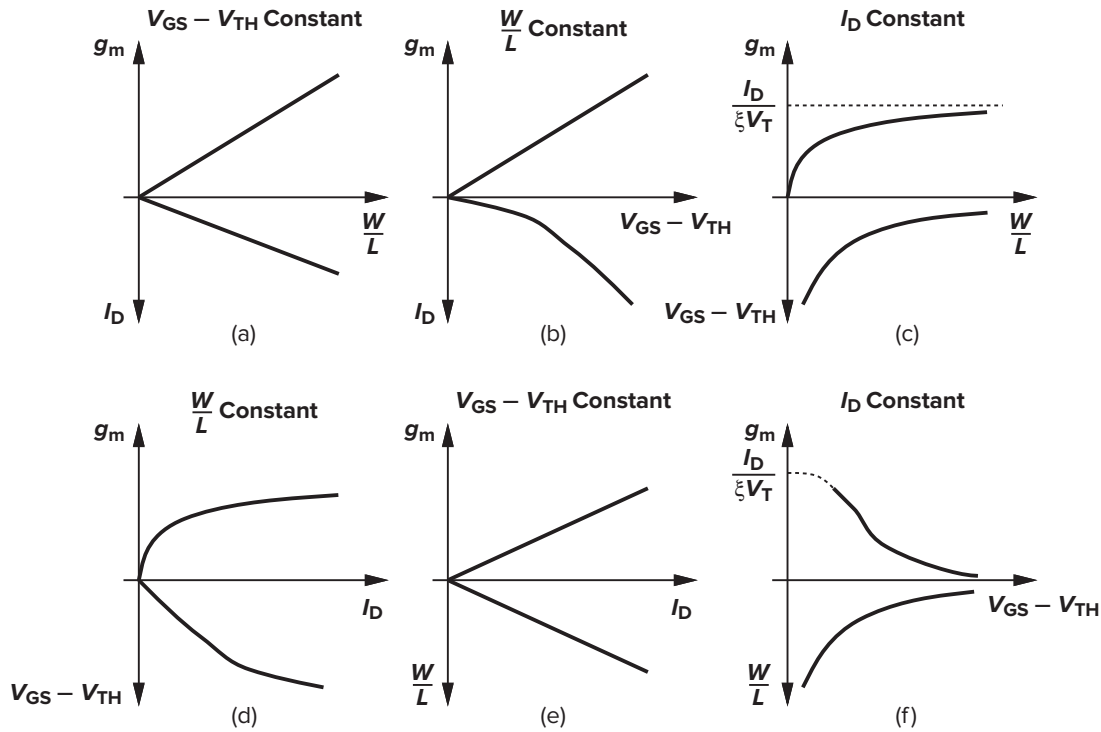


Figure 11.7 Dependence of (a) g_m and I_D upon W/L, (b) g_m and I_D upon $V_{GS} - V_{TH}$, (c) g_m and $V_{GS} - V_{TH}$ upon W/L, (d) g_m and $V_{GS} - V_{TH}$ upon I_D, (e) g_m and W/L upon I_D, and (f) g_m and W/L upon $V_{GS} - V_{TH}$.

can increase $V_{GS} - V_{TH}$ but keep W/L constant [Fig. 11.7(b)], thus requiring a higher drain current. In the former case, the device capacitances rise whereas in the latter, $V_{DS,min}$ increases. In this chapter, we use the notations $V_{DS,min}$, $V_{D,sat}$, and $V_{GS} - V_{TH}$ interchangeably.

From (11.8), we can raise W/L while I_D is constant [Fig. 11.7(c)], as a result of which $V_{GS} - V_{TH}$ is lowered (why?). Due to subthreshold conduction, however, g_m does not climb indefinitely in this case. If we keep W/L constant and increase I_D [Fig. 11.7(d)], then $V_{GS} - V_{TH}$ and hence $V_{DS,min}$ must rise.

From (11.9), we can increase I_D while $V_{GS} - V_{TH}$ is constant [Fig. 11.7(e)]. This requires that W/L increase. Alternatively, we can lower $V_{GS} - V_{TH}$ and keep I_D constant [Fig. 11.7(f)], which means that W/L must increase. For $V_{GS} - V_{TH} \approx 0$, the device enters the subthreshold region and $g_m \approx I_D/(\xi V_T)$. In both cases, the device capacitances climb.

Let us now reconsider the foregoing six scenarios for nanometer devices. We note that the plots in Fig. 11.7 still hold qualitatively, but the g_m and overdrive equations are more complex. The case of Fig. 11.7(a) is particularly interesting and useful and is studied further in the following example.

▶ Example 11.2

The linear scaling of g_m and I_D with W/L, shown in Fig. 11.7(a), holds *regardless* of the transistor characteristics. Explain why.

Solution

Consider, as an example, two identical transistors connected in parallel (Fig. 11.8), each having a transconductance of g_m. If V_{GS} changes by ΔV, then the drain current of each device changes by $g_m \Delta V$, and hence the current of the composite device changes by $2g_m \Delta V$. That is, the parallel combination exhibits a transconductance of $2g_m$. We conclude that increasing *both* the width and the drain current of the transistor by a factor of K (> 1) is equivalent to placing K transistors in parallel and raising the g_m by a factor of K. We say that the scaling preserves the device "current density" (I_D/W) in this case. Note that the bias overdrive voltage remains constant in this scenario, and so does the g_m/I_D ratio. The latter property proves useful in our studies.

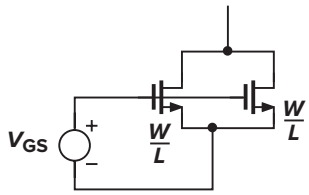


Figure 11.8

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Of the six scenarios depicted in Fig. 11.7, which ones are more common in practice? Since modern analog circuits must operate with low supplies (around 1 V), we often limit $V_{GS} - V_{TH}$ to a few hundred millivolts. Thus, to obtain a certain transconductance, we first keep increasing the width [Fig. 11.7(c)] to the extent that it raises the g_m significantly. As g_m approaches a constant value (in the subthreshold region), the width is no longer a determining factor, leaving the drain current as the only parameter that can increase the g_m [Fig. 11.7(d)]. However, as we increase I_D in this case, $V_{GS} - V_{TH}$ may exceed the allotted value, forcing us to resort to the scenario in Fig. 11.7(e) [which is equivalent to that in Fig. 11.7(a)]. These trials and errors seem rather haphazard, but do not despair! The remainder of this section is dedicated to developing a methodical approach to transistor design. We begin with an important example.

▶ Example 11.3

A transistor having an aspect ratio of $(W/L)_{REF}$ exhibits the g_m-I_D characteristic shown in Fig. 11.9(a).

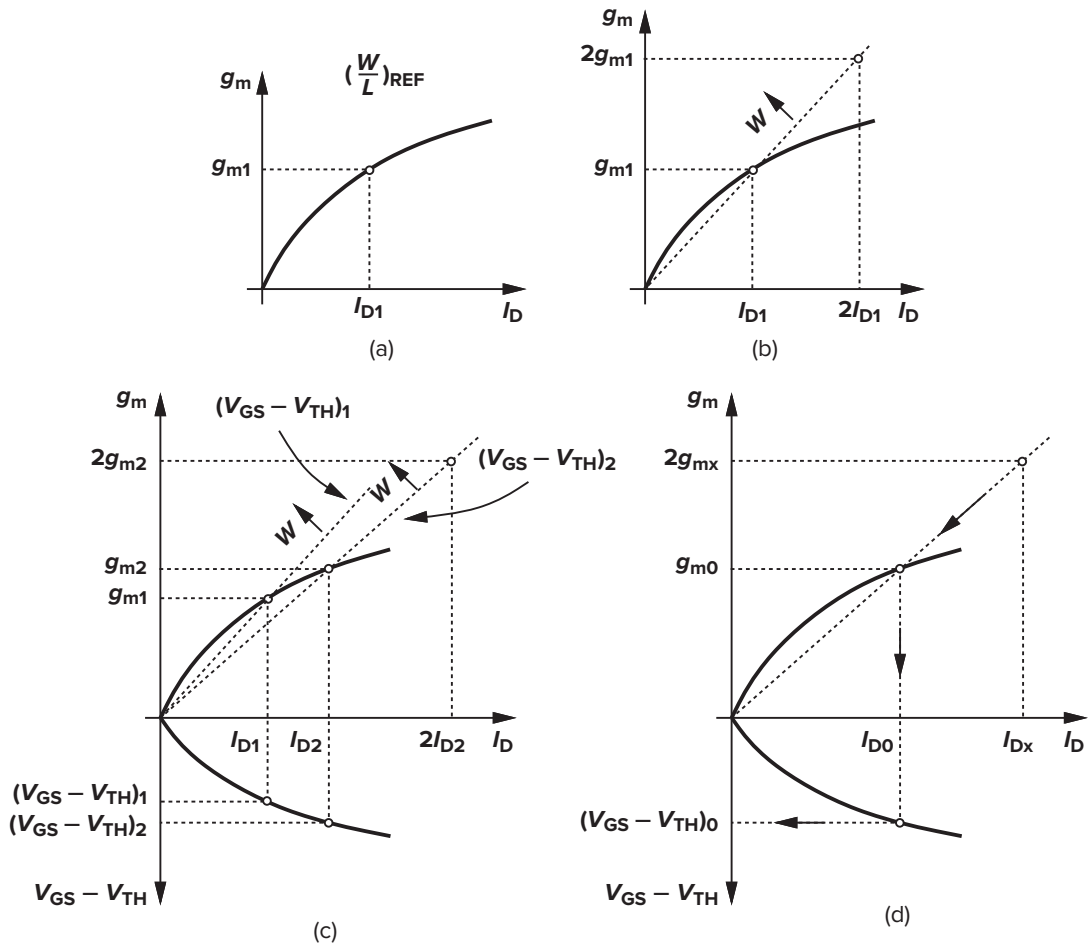


Figure 11.9

(I) Suppose the device is first biased at $I_D = I_{D1}$. What happens to the transconductance and the drain current if the width is doubled while $V_{GS} - V_{TH}$ remains constant? (II) Repeat (I) if we begin with a greater overdrive. (III) We wish to obtain a transconductance of g_{mx} at a drain current of I_{Dx}. How should the transistor be scaled?

Solution

(I) With a constant $V_{GS} - V_{TH}$, doubling the width also doubles the transconductance and the drain current (Example 11.2). Since g_m/I_D is constant, to obtain this point on the g_m-I_D plane, we pass a straight line through the origin and (I_{D1}, g_{m1}), continuing to reach $(2I_{D1}, 2g_{m1})$ [Fig. 11.9(b)]. Thus, all (I_D, g_m) combinations resulting from the scaling of W fall on this line if the overdrive is fixed.

(II) If we begin with a greater overdrive, $(V_{GS} - V_{TH})_2$, the (I_D, g_m) point is located elsewhere, at (I_{D2}, g_{m2}), on the characteristic [Fig. 11.9(c)]. We again draw a straight line through the origin and (I_{D2}, g_{m2}) and continue to $(2I_{D2}, 2g_{m2})$. Thus, each such line in the g_m-I_D plane represents the possible (I_D, g_m) combinations that can be obtained by scaling W for a given overdrive.

(III) We draw a line through the origin and the point (I_{Dx}, g_{mx}) [Fig. 11.9(d)]. The intersection of the line and the g_m plot yields a "reference" point specifying the proper overdrive voltage, $(V_{GS} - V_{TH})_0$, and an acceptable (I_D, g_m) combination, (I_{D0}, g_{m0}). If the width is scaled up by a factor of g_{mx}/g_{m0} $(= I_{Dx}/I_{D0})$, and the overdrive remains equal to $(V_{GS} - V_{TH})_0$, then the desired transconductance and current are obtained.

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11.4 ■ Transistor Design

The reader may have noticed by now that a given transistor in a circuit is characterized by a multitude of parameters. In this section, we assume that transistors operate in saturation and focus on two bias quantities, I_D and $V_{GS} - V_{TH}$ ($= V_{DS,min}$), one small-signal parameter, g_m, and one physical parameter, W/L. A typical transistor design problem specifies two of the first three and seeks the other two (Table 11.1). We wish to develop methodical approaches to computing these two parameters for nanometer devices. While not listed here explicitly, the output resistance, r_O, also proves important in many circuits and is eventually included in our studies in Sec. 11.4.5.

Table 11.1 Three scenarios encountered in transistor design.

	Case I	Case II	Case III
Given	I_D, $V_{DS,min}$	g_m, I_D	g_m, $V_{DS,min}$
To Be Determined	$\frac{W}{L}$, g_m	$\frac{W}{L}$, $V_{DS,min}$	$\frac{W}{L}$, I_D
Design Revision	g_m insufficient; Raise I_D and $\frac{W}{L}$	$V_{DS,min}$ too large; Raise $\frac{W}{L}$	I_D too large; Raise $\frac{W}{L}$; Lower $V_{GS} - V_{TH}$

The reader may recognize that the design problems shown in Table 11.1 are "overconstrained," i.e., the two given parameters inevitably lead to certain values for the other two—even though the results may not always be *desirable*. For example, a known I_D and $V_{DS,min}$ directly give a value for g_m that may not be sufficient for a particular circuit. In such a case, we must modify the design as prescribed in the last row of the table. We will elaborate on this row in the design studies to be followed, but let us make some preliminary remarks here. In Case I, an insufficient g_m would require a higher I_D (possibly exceeding a power budget) and a greater W/L (to satisfy the specified $V_{DS,min}$). In Case II, the given I_D and g_m may yield an unacceptably large $V_{DS,min}$, thereby dictating a greater W/L. In Case III, the necessary I_D may be excessive, demanding a greater W/L and a smaller overdrive.[2]

11.4.1 Design for Given I_D and $V_{DS,min}$

A common situation that arises in analog design is as follows. For a particular transistor in the circuit, we have chosen a bias current (perhaps according to a power budget) and a minimum V_{DS} (perhaps according to the voltage headroom, i.e., the restrictions imposed by the supply voltage and the required swings).[3] We now wish to determine the dimensions and the transconductance of the device, recognizing that the square-law equations are inaccurate. Of course, with the transistor models available, we can simulate the device and obtain these values, but we seek a more methodical and less laborious procedure. Our approach proceeds in three steps. We consider $I_D = 0.5$ mA and $V_{DS,min} = 200$ mV as an example.

Step 1 Select a "reference" transistor, with a width W_{REF} and a length equal to the minimum allowable value, L_{min} (e.g., $L_{min} = 40$ nm). Let us choose $W_{REF} = 5\ \mu$m as an example.

Step 2 Using the actual device models and a circuit simulator, plot the I_D-V_{DS} characteristics of the reference transistor for different values of $V_{GS} - V_{TH}$. In typical analog circuits, $V_{GS} - V_{TH}$ ranges from about 50 mV to about 600 mV. We can therefore construct the characteristics with the overdrive

[2] In cases I and III, raising W/L can relax the g_m-I_D trade-off only if the device does not enter the subthreshold region.

[3] We assume that the supply voltage is given.

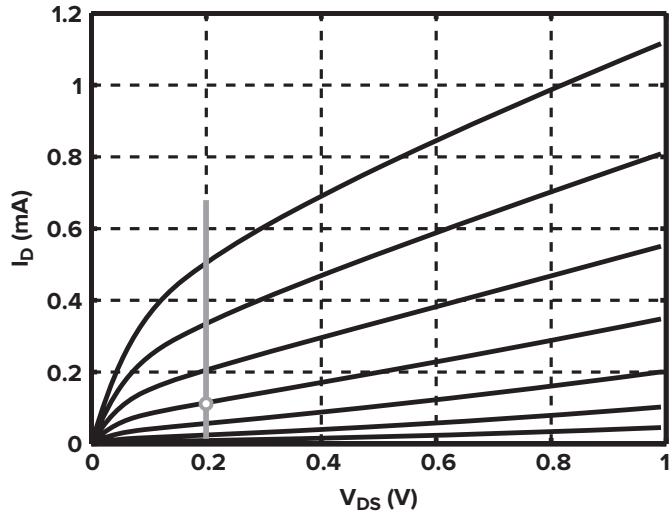


Figure 11.10 Drain current for $V_{GS} - V_{TH} = 50$ mV $\cdots$ 350 mV in steps of 50 mV for a reference device.

incrementing in steps of 50 mV.[4] Figure 11.10 shows the results for $W_{REF}/L_{min} = 5\ \mu$m/40 nm. (Here, $V_{GS} - V_{TH}$ increments from 50 mV to 350 mV for clarity.)

Step 3 Bearing in mind that our example specifies $I_D = 0.5$ mA and $V_{DS,min} = 200$ mV, we draw a vertical line at $V_{DS} = 200$ mV (Fig. 11.10) and find its intersection with the plots. Which plot should we select? If the device obeyed the square law, we would choose the plot for $V_{GS}-V_{TH} = V_{DS,min} = 200$ mV. However, the short-channel device remains in saturation even for $V_{GS}-V_{TH} = 350$ mV at $V_{DS} = 200$ mV. The situation is therefore more complex, but let us proceed with $V_{GS} - V_{TH} = 200$ mV for now.

Step 4 The foregoing procedure has yielded, for the reference transistor, one operating point that satisfies the V_{DS} requirement. The drain current, $I_{D,REF}$, however, may not be close to the necessary value, 0.5 mA in our example. What shall we do here? We must now *scale* the width of the transistor and hence its drain current. Since in Fig. 11.10, $I_{D,REF} \approx 100\ \mu$A, we choose a transistor width of (500 μA/100 μA)$\times W_{REF} = 5W_{REF} = 25\ \mu$m.

How much is the transconductance of the earlier reference transistor? We recognize from the plots of Fig. 11.10 that, as $V_{GS}-V_{TH}$ is incremented from 200 mV to 250 mV, I_D changes by about 100 μA. Thus, $g_m \approx 100\ \mu$A/50 mV$= 2$ mS. Since the change in the overdrive is not much less than the initial value of 200 mV, we may seek a more accurate value for g_m. To this end, let us return to the reference transistor and, using simulations, plot its transconductance as a function of $V_{GS} - V_{TH}$ with $V_{DS} = 200$ mV. For a square-law device, this plot would be a straight line, $g_m = \mu_n C_{ox}(W_{REF}/L_{min})(V_{GS} - V_{TH})$, but with short-channel effects, g_m eventually saturates. Shown in Fig. 11.11, the result predicts $g_m = 1.5$ mS for $V_{GS} - V_{TH} = 200$ mV. Now, if both the width and the drain current are scaled up by a factor of 5, then g_m also rises by the same factor (Example 11.2), reaching a value of 7.5 mS. As indicated in Table 11.1, if this transconductance is insufficient, W/L must be increased further.

With the I_D and g_m plots obtained for the reference device, we can readily perform scaling to determine the width and transconductance of other transistors in a circuit. The key point here is that the I_D and g_m simulations are performed only once (for a given channel length) but serve most of our design work.

[4] Our approach deals with only moderate and strong inversion, as is the case in most analog circuits.

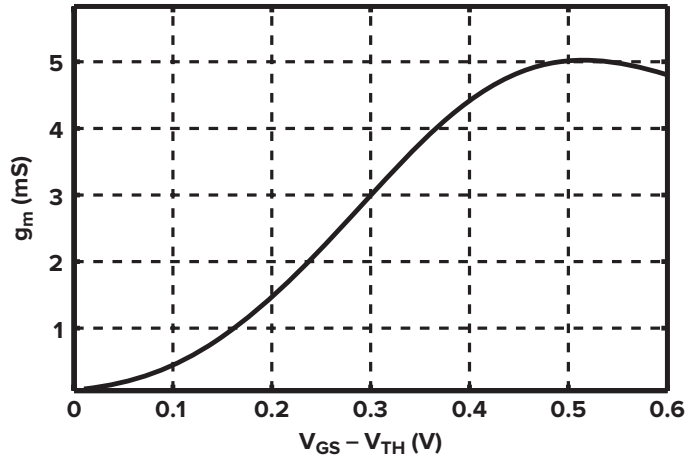


Figure 11.11 Dependence of g_m on overdrive for $W/L = 5\ \mu$m/40 nm and $V_{DS} = 200$ mV.

Can we choose a higher overdrive voltage in Fig. 11.10? Suppose we select $V_{GS} - V_{TH} = 250$ mV, obtaining $I_D = 200\ \mu$A for the reference transistor and a transconductance of about 2.3 mS from Fig. 11.11. If scaled up to 12.5 μm so as to carry 500 μA, the transistor exhibits a transconductance of 2.5×2.3 mS $= 5.75$ mS, a value *less* than that observed in the previous case (7.5 mS). This occurs because $g_m = 2I_D/(V_{GS} - V_{TH})$ in saturation. To obtain a high transconductance, therefore, we typically choose $V_{GS} - V_{TH} \approx V_{DS,min}$ even though it translates to a wider transistor.

▶ Example 11.4

The circuit shown in Fig. 11.12 must be designed for a power budget of 1 mW and a peak-to-peak output voltage swing of 0.8 V. Assuming $L = 40$ nm for M_1, compute its required width. Can the transistor provide a transconductance of $1/(50\ \Omega)$?

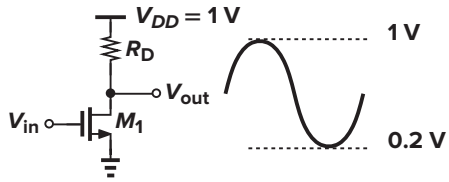


Figure 11.12

Solution

The power budget along with $V_{DD} = 1$ V translates to a drain bias current of 1 mA. For the circuit to accommodate an output swing of 0.8 V, M_1 must remain in saturation as V_{DS} falls to 0.2 V. We return to the I_D-V_{DS} characteristics of Fig. 11.10 and recall that $I_{D,REF} \approx 100\ \mu$A for $V_{DS} = V_{GS} - V_{TH} = 200$ mV. We must therefore scale W_{REF} up by a factor of 1 mA/0.1 mA, obtaining $W/L = 50\ \mu$m/40 nm. The transconductance is also multiplied by this factor, reaching 15 mS $= 1/(67\ \Omega)$. Note that these results are independent of the value of R_D.

We conclude that, if the transistor is designed simply to satisfy this example's I_D and V_{DS} specifications, then it does not necessarily achieve a transconductance of $1/(50\ \Omega)$.

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In addition to I_D, $V_{GS} - V_{TH}$, and g_m, the output impedance of the transistors also becomes important in many analog circuits. As explained in Chapter 17, r_O cannot be expressed as $1/(\lambda I_D)$ for short-channel devices. The value of r_O can be estimated from the slope of the I_D characteristics in Fig. 11.2, but for convenience and accuracy, we use simulations to plot r_O for the reference transistor as a function of I_D (Fig. 11.13).

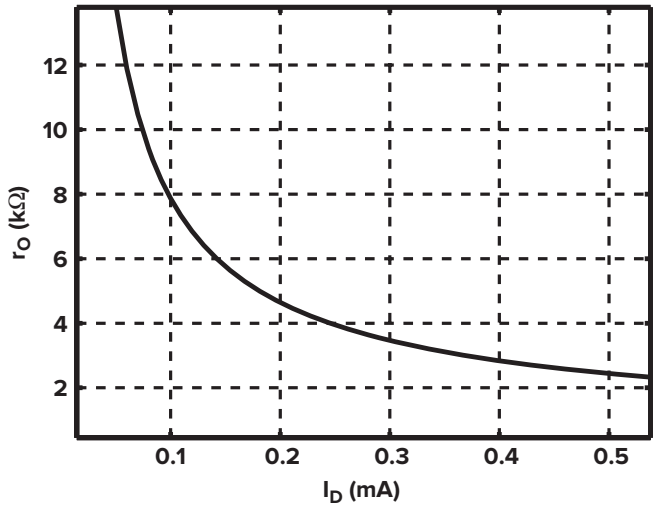


Figure 11.13 Output resistance of a 5-μm/40-nm NMOS device as a function of drain current.

▶ **Example 11.5**

Determine the output resistance of M_1 in Example 11.4.

Solution

The reference transistor in Example 11.4 carries a current of 100 μA, exhibiting an output resistance of 8 kΩ. Since both the width and the drain current are scaled up by a factor of 10, the output resistance drops by the same factor, falling to 800 Ω.

◀

11.4.2 Design for Given g_m and I_D

In many analog circuits, a given transistor must provide sufficient transconductance while consuming minimal power. We thus begin with a specified transconductance, g_{m1}, and an upper limit for the drain bias current, I_{D1}, seek the corresponding values of W/L and $V_{GS} - V_{TH}$. In this section, we assume that $g_{m1} = 10$ mS and $I_{D1} = 1$ mA. Of course, our first task is to determine whether g_{m1} can be obtained at all with $I_D \leq I_{D1}$. The maximum g_m occurs in the subthreshold region (if W/L is large) and is given by $I_D/(\xi V_T)$, where $\xi \approx 1.5$ (Chapter 2). For example, if $I_D = 1$ mA, then g_m cannot exceed 26 mS at the room temperature.

Since in our example, $g_{m1} < I_{D1}/(\xi V_T)$, we can proceed to design the transistor. The reader is encouraged to first read Example 11.3 carefully.

Step 1 Using simulations, we plot g_m as a function of I_D for a reference transistor, e.g., with $W_{REF}/L_{min} = 5$ μm/40 nm (Fig. 11.14).

Step 2 We identify the point (I_{D1}, g_{m1}) on the g_m-I_D plane and draw a line through the origin and this point, obtaining the intersection at $(I_{D,REF}, g_{m,REF}) = (240\ \mu\text{A}, 2.4\ \text{mS})$ and a corresponding overdrive.

Step 3 We multiply W_{REF} by $g_{m1}/g_{m,REF} = 4.2$ so as to travel on the straight line to point (I_{D1}, g_{m1}) while maintaining the same overdrive (Example 11.3). This completes the design of the transistor.

The above procedure elicits two questions. First, does the straight line passing through the origin and (I_{D1}, g_{m1}) always intersect the g_m-I_D plot? If we consider a square-law device in strong inversion, then $g_m = \sqrt{2\mu_n C_{ox}(W/L)I_D}$ has a slope of infinity at the origin, guaranteeing an intersection point. In the

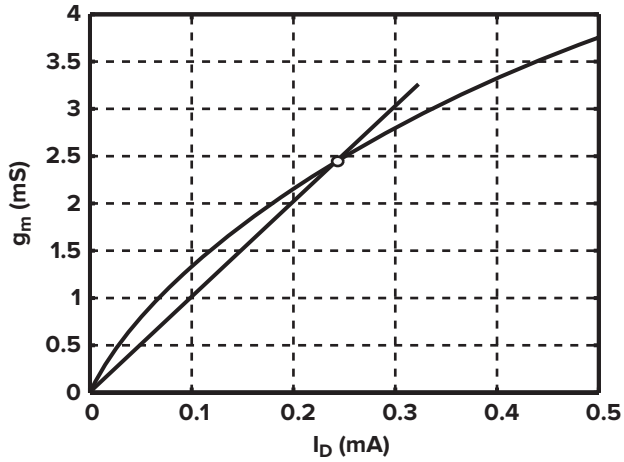


Figure 11.14 Transconductance as a function of I_D for $W/L = 5\ \mu$m/40 nm.

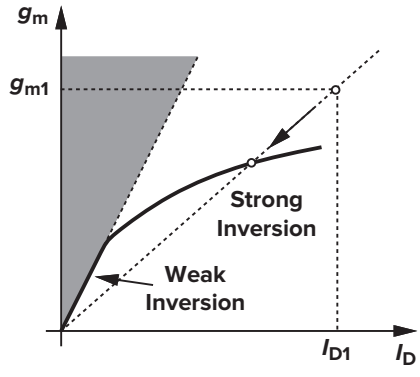


Figure 11.15 Unachievable g_m region.

subthreshold region, on the other hand, $g_m \propto I_D$ (Fig. 11.15), which means that the (I_D, g_m) combinations in the gray region are not achievable.

The second question is, what if $(V_{GS} - V_{TH})_{REF}$ is excessively large? As stipulated in Table 11.1, we must then increase W further, but by what factor? Suppose, as shown in Fig. 11.16, an overdrive of $(V_{GS} - V_{TH})_2 < (V_{GS} - V_{TH})_{REF}$ is desired. We then find the corresponding current, I_{D2}, and transconductance, g_{m2}, on the g_m-I_D plane. Next, we draw a line through the origin and the point (I_{D2}, g_{m2}) and continue to $I_D = I_{D1}$, i.e., we multiply W_{REF} by I_{D1}/I_{D2}. The resulting width guarantees an overdrive of $(V_{GS} - V_{TH})_2$ at a drain current of I_{D1} and provides a transconductance of *at least* g_{m1}. The new transconductance, g'_{m1}, is inevitably greater because the width has been increased beyond $g_{m1}/g_{m,REF}$ $(= I_{D1}/I_{D,REF})$.

11.4.3 Design for Given g_m and $V_{DS,min}$

In some designs, the transconductance is dictated by some performance requirements (voltage gain, noise, etc.) and the minimum V_{DS} by the voltage headroom—with no explicit specification of I_D. Of course, each circuit eventually faces a power budget and hence an upper bound on its bias current(s).

The design procedure for obtaining a transconductance of g_{m1} at $V_{DS,min}$ in this case is as follows.

Step 1 We use simulations to plot the g_m as a function of $V_{GS} - V_{TH}$ for the reference transistor (Fig. 11.17). Now, we select $(V_{GS} - V_{TH})_1 = V_{DS,min}$ and obtain the corresponding transconductance, $g_{m,REF}$. In this case, it is helpful to plot I_D on the same plane and find $I_{D,REF}$ at $(V_{GS} - V_{TH})_1$.

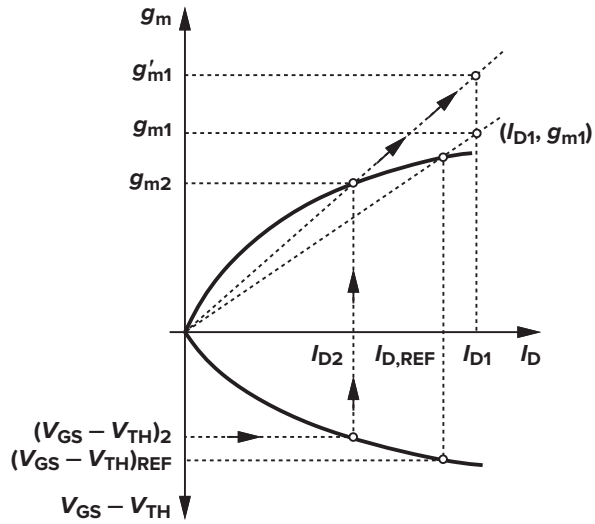


Figure 11.16 Modification of transistor design for a lower overdrive.

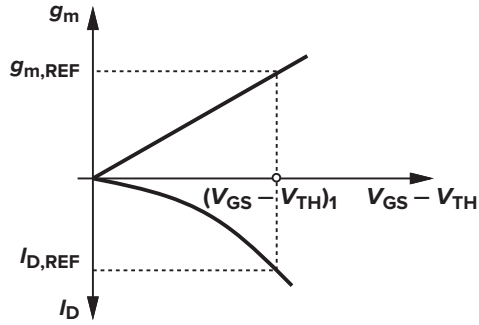


Figure 11.17 Calculation of $g_{m,REF}$ for a given overdrive.

Step 2 To reach the required transconductance, g_{m1}, we scale the transistor width up by a factor of $g_{m1}/g_{m,REF}$. Note that I_D scales by the same factor.

These two steps complete the design, but what if the resulting I_D is excessively large? We can return to Case II in Sec. 11.4.2 and redesign for a given g_m and I_D. The device is now wider and has a smaller transconductance.

As can be seen from our procedures in this section, we have portrayed the overdrive voltage (or $V_{D,sat}$) as an indispensable dimension in our device design. This is because today's low supply voltages have made the problem of headroom more severe than ever.

11.4.4 Design for a Given g_m

Our approach has assumed that the drain current and the overdrive voltage are specified and the other device parameters must be determined. Since power consumption and voltage headroom prove critical in today's analog design, this assumption holds in most cases. However, suppose a design problem specifies only the transconductance, and we wish to compute the remaining parameters. How do we select the transistor's drain current, overdrive voltage, and dimensions?

Two scenarios must be envisaged. (1) We select a certain W/L and raise I_D until we obtain the desired transconductance, g_{m1}. In this case, the required I_D, and hence the power consumption, may be excessive. More important, the overdrive voltage may be unacceptably large, leaving little headroom for voltage swings. (2) We select a reasonable value for I_D (perhaps according to a power budget) and

increase W/L to obtain g_{m1}. In this case, however, we may *not* be able to reach g_{m1}; increasing W/L (and hence decreasing V_{GS}) eventually drives the device into the subthreshold region, where g_m cannot exceed $I_D/(\xi V_T)$. This means that the selected current is insufficient, i.e., we should always briefly check to see if the upper limit given by $I_D/(\xi V_T)$ can be met with the current budget.

The above scenarios indicate the need for a systematic approach to the selection of device parameters when only g_{m1} is known. To this end, we return to the concept of the reference device and, using simulations, construct two plots for it. Shown in Fig. 11.18, the two represent g_m and $V_{GS} - V_{TH}$ as a function of I_D.[5] We begin by selecting a reasonable value for $V_{GS} - V_{TH}$, e.g., 200 mV, which points to $I_{D,REF}$ and $g_{m,REF}$. Now, we scale the width and the drain current by a factor of $g_{m1}/g_{m,REF}$.

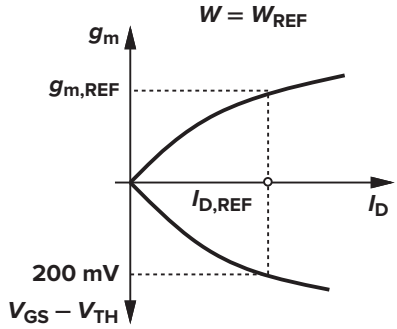


Figure 11.18 Translation of overdrive to $g_{m,REF}$.

What if the foregoing method yields an unacceptably high I_D? We can choose a smaller overdrive, e.g., 150 mV, and repeat the earlier steps.

11.4.5 Choice of Channel Length

If the selection of I_D, $V_{GS} - V_{TH}$, and g_m does not yield a sufficiently high r_O, we must increase the length of the transistor. Of course, to maintain the same drain current, overdrive voltage, and g_m, the *width* must also be increased proportionally. However, such scaling of the length and the width is not straightforward because, as the *drawn* length is increased from L_{min} to, say, $2L_{min}$, the effective length rises from $L_{min} - 2L_D$ to $2L_{min} - 2L_D$, i.e., by a factor of less than 2. For this reason, we must use simulations to construct the I_D-V_{DS}, g_m and r_O characteristics for several channel lengths, e.g., 60 nm, 80 nm, and 100 nm (drawn values).

11.5 ■ Op Amp Design Examples

In this section, we wish to repeat the op amp design examples studied in Chapter 9 in 40-nm technology. We target the following typical specifications:

- Differential Output Voltage Swing = 1 V_{pp}
- Power Consumption = 2 mW
- Voltage Gain = 500
- Supply Voltage = 1 V

[5]Here, V_{DS} is constant and approximately equal to $V_{DD}/2$. In nanometer technologies, different V_{DS} values alter these characteristics to some extent.

The single-ended output swing of 0.5 V_{pp} is small enough to make telescopic or folded-cascode op amps a plausible choice. We therefore explore these topologies before deciding whether a two-stage op amp is necessary.

A few notes about our transistor sizing methodology are warranted. We wish to begin with the minimum allowable width and length unless otherwise dictated by current, transconductance, $V_{D,sat}$, output resistance, or other requirements. Interestingly, in the designs pursued in this chapter, all transistor widths are greater than the minimum value. Also, for simplicity, we may scale the drawn W and L by the same factor even though the effective W/L does not remain exactly constant.

11.5.1 Telescopic Op Amp

Can a telescopic-cascode op amp topology meet the above specifications? In this section, we explore this possibility. It may not, but we will learn a great deal. Consider the circuit shown in Fig. 11.19. Of the total supply current of 2 mA, we allow 50 μA for I_{REF1}, 50 μA for I_{REF2}, and 0.95 mA for each branch of the differential pair. We must now allocate the transistor drain-source voltages so as to accommodate a single-ended peak-to-peak output swing of 0.5 V; i.e., we must distribute the remaining 0.5 V over M_9, $M_{1,2}$, $M_{3,4}$, $M_{5,6}$, and $M_{7,8}$. Let us allow a V_{DS} of 100 mV for each—even though the PMOS devices have a lower mobility. With the bias currents and overdrives known, we can determine the W/L's by examining the transistor I/V characteristics.

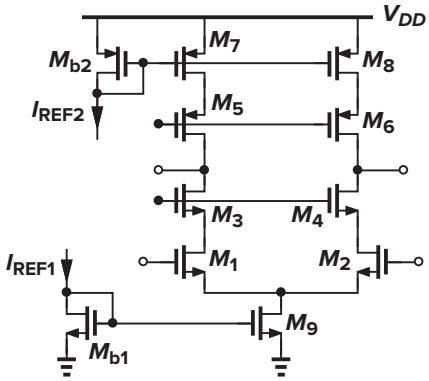


Figure 11.19 Telescopic-cascode op amp.

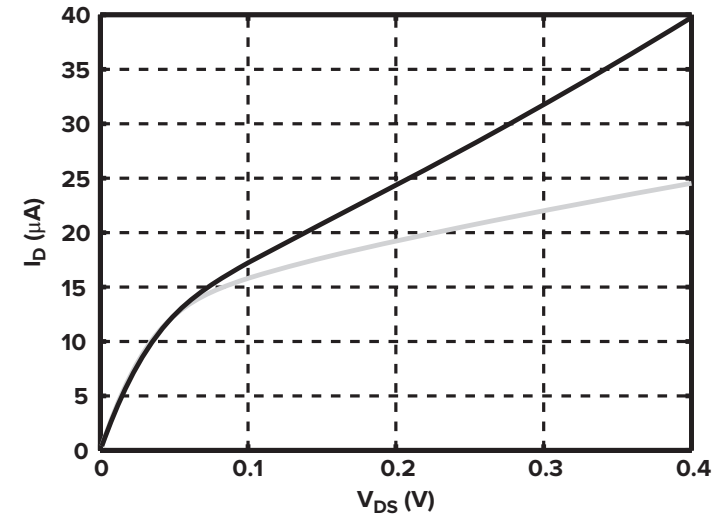
Before delving into details, however, we should pause and think about the feasibility of the design, specifically in terms of the required voltage gain. We make three observations: (1) for $L = 40$ nm, the intrinsic gain, $g_m r_O$, of NMOS devices is around 7 to 10 and that of PMOS devices around 5 to 7, (2) for reasonable device dimensions, it is difficult to raise $g_m r_O$ beyond 10 for PFETs (unless we allow longer lengths and hence lower speeds), and (3) if we approximate g_m as $2I_D/(V_{GS} - V_{TH}) = 2 \times 0.95$ mA/100 mV = 19 mS, we estimate $r_O \approx 530\ \Omega$ from $g_m r_O \approx 10$.

Let us now apply the foregoing values to the telescopic topology in Fig. 11.19. If $g_{m1,2} \approx 19$ mS, then for the gain, $G_m R_{out}$, to reach 500, the op amp output impedance must exceed 26 kΩ. equal to $(g_{m5,6} r_{O5,6}) r_{O7,8}$, pointing to a serious limitation. However, with $g_{m3,4} r_{O3,4} \approx 10$ and $r_{O7,8} \approx 530\ \Omega$ (from the third observation above), we have $(g_{m3,4} r_{O3,4}) r_{O1,2} \approx 5.3$ kΩ, obtaining a voltage gain of only about 100 even if the PMOS devices have $\lambda = 0$! This fivefold deficit makes the telescopic arrangement impractical for a gain of 500.

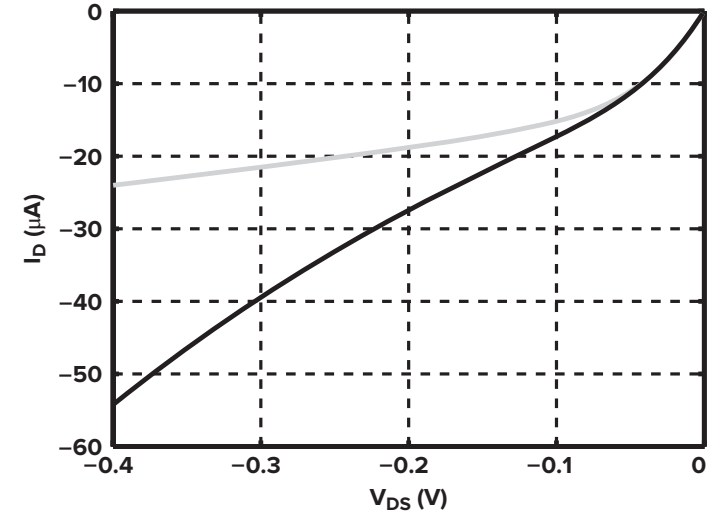
Out of curiosity, we still continue with the design and see what performance we can achieve. To this end, we use simulations to construct the I/V characteristics of NMOS and PMOS devices with $L = 40$ nm and 80 nm, predicting that the minimum length exhibits an unacceptably low r_O and $g_m r_O$. The simulation parameters must also ensure that the devices remain in saturation for $|V_{DS}| \geq 100$ mV. Given that the

threshold and the overdrive elude a clear definition in nanometer technologies, we must adjust V_{GS} in simulations to ensure saturation.

The results are plotted in Fig. 11.20(a) for $(W/L)_N = 5\ \mu$m/40 nm and 10 μm/80 nm with $V_{GS} =$ 300 mV, and in Fig. 11.20(b) for $(W/L)_P = 5\ \mu$m/40 nm and 5 μm/80 nm with $V_{GS} = -400$ mV.[6] We should make some remarks. First, it is difficult to distinguish between triode and saturation regions, especially for PFETs. In fact, the 40-nm PMOS device behaves almost as a resistor and displays a



(a)



(b)

Figure 11.20 I_D-V_{DS} characteristics for (a) an NMOS device with $V_{GS} = 300$ mV and $W/L = 5\ \mu$m/40 nm (black plot) or 10 μm/80 nm (gray plot), and (b) a PMOS device with $V_{GS} = -400$ mV and $W/L = 5\ \mu$m/40 nm (black plot) or 5 μm/80 nm (gray plot).

[6]The width of the PMOS device is not scaled here to reveal the small change in I_D as the drawn L is doubled. This occurs because V_{TH} *falls* as L increases from its minimum value (Chapter 17).

decreasing output impedance as $|V_{DS}|$ approaches 400 mV.[7] For the other three characteristics, we can roughly identify a "knee" point beyond which the slope falls considerably. The gate-source voltages have been chosen to place this point below $|V_{DS}| = 100$ mV.

Second, at $V_{DS} = 100$ mV, the 10-μm/80-nm NMOS transistor provides an output resistance of 22.8 kΩ and a drain current of 16 μA. If scaled up to carry 950 μA with the same terminal voltages, the device exhibits an r_O of 385 Ω! Similarly, the 5-μm/80-nm PMOS transistor has an r_O of 18.45 kΩ at $V_{DS} = -100$ mV with $I_D = 15$ μA, thus offering $r_O = 290$ Ω if scaled up to carry 950 μA. These very low r_O values are quite discouraging, but we will continue to explore.

We now scale the NMOS and PMOS device widths to accommodate a drain current of 950 μA with $V_{GS,N} = 300$ mV, $V_{GS,P} = -400$ mV, and $|V_{DS}| = 100$ mV. The resulting design is shown in Fig. 11.21.[8] As a general principle, we prefer to use minimum-length devices in the signal path so as to maximize their speed (or at least minimize their capacitances for a given g_m). It is surprising to see such large widths in 40-nm technology for a drain current of 950 μA, an inevitable outcome of confining $|V_{DS}|$ to 100 mV.

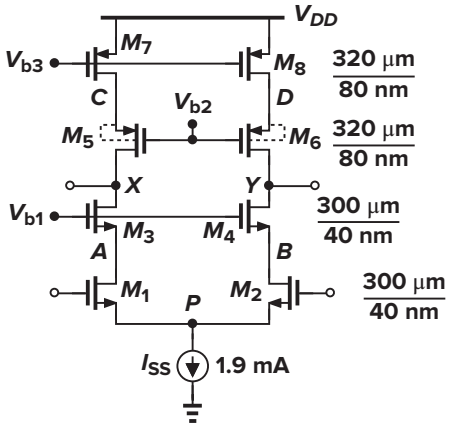


Figure 11.21 First design of telescopic-cascode op amp.

The bias voltages are tentatively chosen as follows: (a) the input common-mode level, $V_{CM,in}$, is equal to 100 mV for the tail current source plus $V_{GS1,2}(= 300$ mV), (b) V_{b1} is equal to $V_{D1,2}$ $(= 200$ mV) plus $V_{GS3,4}(= 300$ mV), (c) V_{b2} is equal to $V_{DD} - |V_{DS7,8}| - |V_{GS5,6}|$, and (d) V_{b3} is equal to $V_{DD} - |V_{GS7,8}|$. Upon simulating the circuit with these values, the reader may encounter a very low or high output common-mode level. This effect arises from the absence of common-mode feedback and hence the departure of $|I_{D7,8}|$ from 1.9 mA/2. We avoid this issue by a slight adjustment of V_{b3} for now.

We perform a dc sweep simulation of the differential input voltage, V_{in}, and examine the voltages at various nodes in Fig. 11.21 to ensure that the transistors are "healthy." Plotted in Fig. 11.22, the drain voltages of M_1 and M_2 are around 220 mV in the middle of the range. Similarly, the drain voltages of M_7 and M_8 are close to the targeted value.

Next, we study the output behavior, depicted in Fig. 11.22 by V_X and V_Y. The slope of each single-ended output is approximately equal to 15 in the vicinity of $V_{in} = 0$, yielding a differential gain of 30, far below our target. Can this design deliver a single-ended peak-to-peak swing of 0.5 V? We note that the characterisitic becomes very nonlinear as each output rises toward 0.7 V. In fact, around this output level, the slope gives a differential gain of about 6.4.

[7] As explained in Chapter 17, the fall in the output impedance arises from drain-induced barrier lowering (DIBL).

[8] The bulks of M_5 and M_6 are tied to their respective sources so as to remove body effect. While not essential, this arrangement reduces $|V_{GS5,6}|$, providing more comfortable margins in the design.

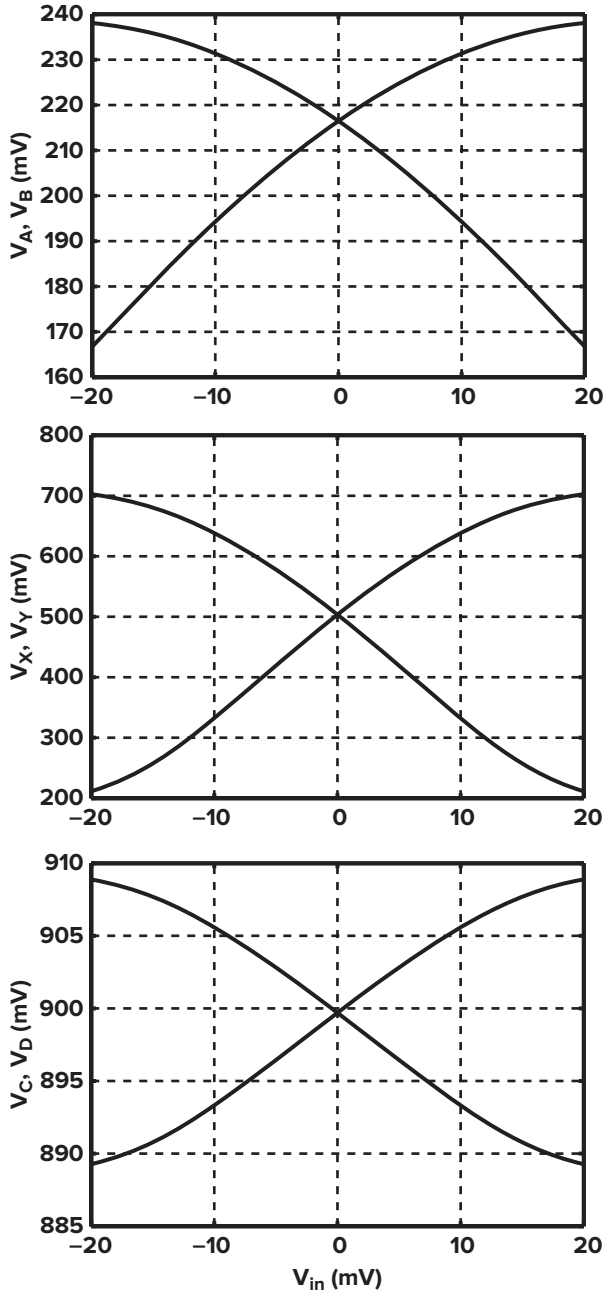


Figure 11.22 Behavior of the voltages at the drains of input transistors (V_A, V_B), the output nodes (V_X, V_Y), and the drains of PMOS current sources (V_C, V_D).

▶ **Example 11.6**

The slope of the characteristics in Fig. 11.22 predicts a differential gain of 3 from the input to nodes A and B. Explain the reason for such a high gain at the cascode nodes.

Solution

Recall from Chapter 3 that the impedance seen at the source of a cascode device is roughly equal to the impedance seen at its drain divided by its $g_m r_O$. Due to the low $g_m r_O$, the impedance seen at A and B is quite a lot higher than $1/g_{m3,4}$, leading to a large gain.

◀

Let us raise the gain by increasing $(W/L)_{3,4}$ to 600 μm/80 nm. Plotted in Fig. 11.23, the characteristics exhibit a gain of about 54 but still a limited output swing.

Bias Circuit The op amp of Fig. 11.21 relies on the proper choice of I_{SS}, V_{b1}, V_{b2}, and V_{b3}. We must therefore design a circuit to generate these bias quantities. We recognize that I_{SS} and V_{b3} must be established by current mirror action (why?) and V_{b2} by low-voltage cascode biasing. The bias voltage V_{b1} requires a different approach.

We begin with $I_{SS} = 1.9$ mA, choosing a channel length of 40 nm and, scaling from Fig. 11.20, a width of 600 μm for $V_{DS} = 100$ mV. Utilizing a reference budget current of 25 μA, we arrive at the arrangement shown in Fig. 11.24(a), where W_{12} is scaled down from W_{11} by a factor of 1.9 mA/25 μA. Since M_{11} operates with a V_{DS} of 100 mV, we insert R_1 in series with the drain of M_{12} and select its value such that $V_{DS12} = V_{GS12} - V_{R1} = 100$ mV.

The above bias design is still sensitive to the CM level sensed by M_1 and M_2 because $V_{DS11} = V_{CM,in} - V_{GS1,2}$, whereas $V_{DS12} = V_{GS1,2} - V_{R1}$. In other words, we must ensure that the drain voltage of M_{12} tracks $V_{CM,in}$. This can be accomplished as shown in Fig. 11.24(b), where R_1 is replaced by a differential pair driven by V_{in1} and V_{in2}. With proper scaling of the widths, we now have $V_{GS13,14} = V_{GS1,2}$, and hence $V_{DS12} = V_{DS11}$.

Next, we deal with the generation of V_{b1} in Fig. 11.21. This voltage must be equal to $V_{GS3,4} + V_{DS1,2} + V_P$, where $V_{DS1,2} = 100$ mV. Since V_{b1} is higher than V_P by $V_{GS3,4} + V_{DS1,2}$, we surmise that a diode-connected device in series with a drain-source voltage added to V_P can produce V_{b1}. Illustrated in Fig. 11.25, the idea is to match V_{GS15} to $V_{GS3,4}$ and V_{DS16} to $V_{DS1,2}$. The bias current I_b must be much less than I_{SS} so as to negligibly affect the power budget. We select $I_b = 15$ μA, and hence $(W/L)_{15,16} = 10$ μm/80 nm.[9] It is important to observe how V_{b1} tracks $V_{CM,in}$: if $V_{CM,in}$ goes up, so do V_P and, consequently, V_{b1}, thus keeping $V_{DS1,2}$ constant. That is, M_{15} and M_{16} operate as level shifters. If V_{b1} were *constant*, a rise in $V_{CM,in}$ would inevitably reduce $V_{DS1,2}$ and the gain.

In order to generate V_{b3} and V_{b2}, we construct a low-voltage cascode bias network as shown in Fig. 11.26. Here, transistors M_{17} and M_{18} are scaled down from $M_{7,8}$ and $M_{5,6}$, respectively, ensuring that $V_{DS17} = V_{DS7,8}$. To create $V_{b2} = V_{DD} - |V_{DS7,8}| - |V_{GS5,6}|$, we again employ a diode-connected device, M_{20}, in series with a V_{DS} (produced by M_{19}).

We should emphasize that the very narrow voltage margins dictated by the low supply make this design sensitive to mismatches between the bias branches and the core of the circuit. For example, a mismatch between V_{GS18} and $V_{GS5,6}$ can leave less $|V_{DS}|$ for $M_{7,8}$, pushing these two current sources below the knee point. Also, note that we still have a few ideal current sources, which would be copied from a bandgap reference (Chapter 12).

Common-Mode Feedback With various mismatches present in the above op amp design, the PMOS currents in Fig. 11.21 are not exactly equal to $I_{SS}/2$, forcing the output CM level toward V_{DD} or ground

[9] With the values chosen here, $V_{DS16} < V_{DS1,2}$ because $V_{GS16} > 300$ mV (why?). For this reason, some adjustment in simulations is necessary.

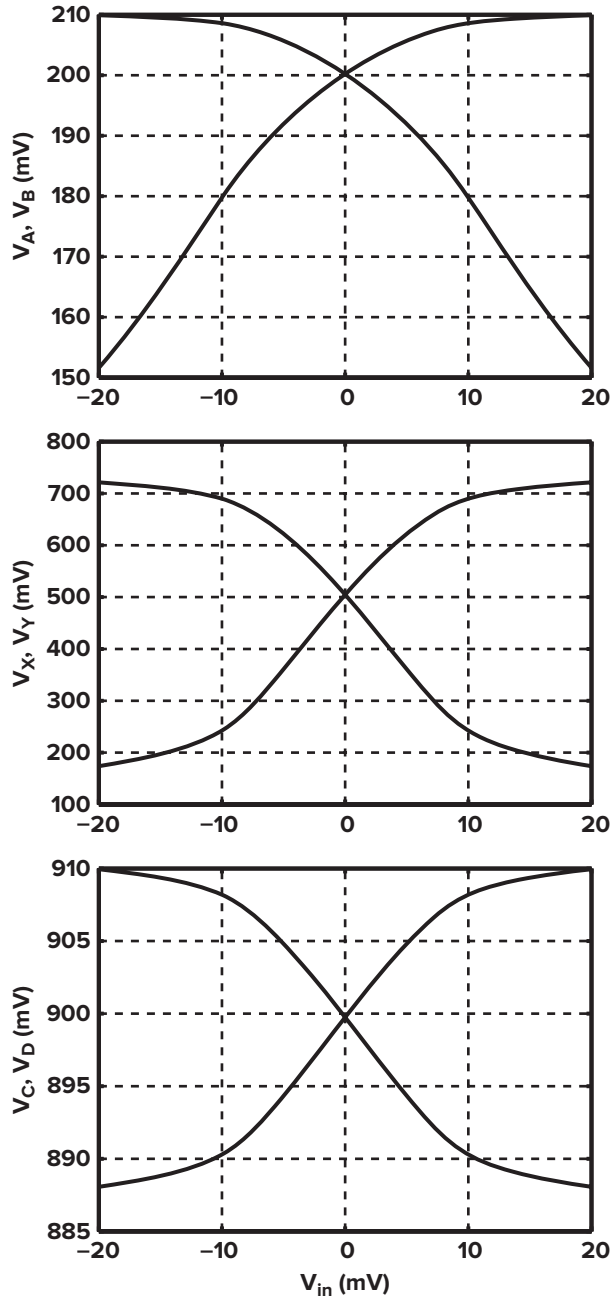


Figure 11.23 Behavior of the voltages at the drains of input transistors (V_A, V_B), the output nodes (V_X, V_Y), and the drains of PMOS current sources (V_C, V_D).

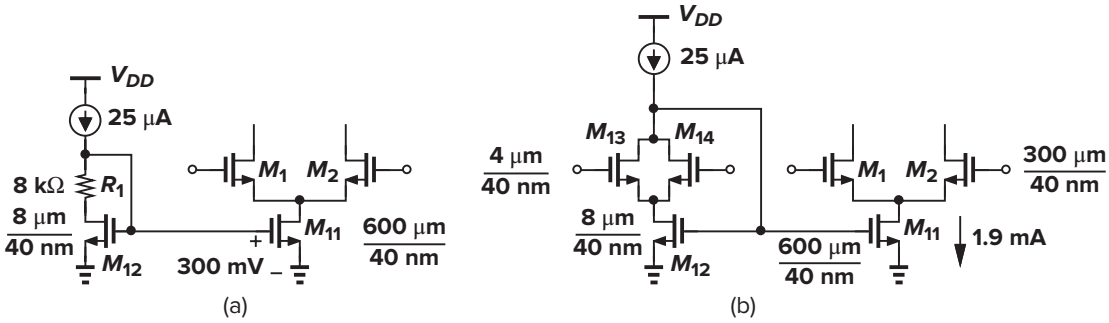


Figure 11.24 (a) Simple and (b) more accurate biasing for tail current source.

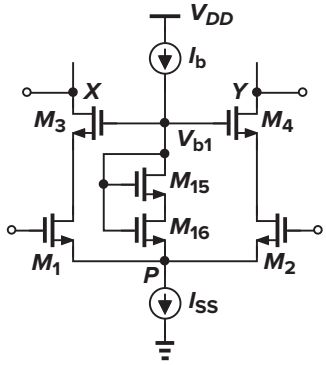


Figure 11.25 Generation of cascode gate bias voltage.

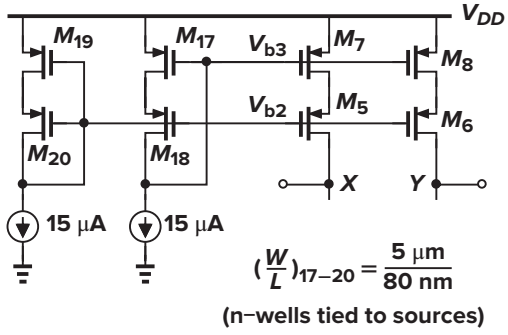


Figure 11.26 Generation of bias for PMOS cascode current sources.

and hence requiring CMFB. We must sense the output CM level, V_{CM}, and feed the result back to the NMOS or PMOS current sources.

Recall from Chapter 9 that the CM level can be sensed by resistors, triode transistors, or source followers. The high output impedance of the op amp dictates very large resistors,[10] and the tight voltage margins demand precise CM control and preclude triode devices. The only solution, source followers, however, cannot measure the CM level across a wide output swing. As shown in Fig. 11.27(a), if V_X (or V_Y) falls (in response to differential signals), I_1 (or I_2) eventually collapses, disabling the source follower. But, is it possible to complement the NMOS followers by PMOS counterparts? Consider the arrangement depicted in Fig. 11.27(b), where PMOS followers M_{23} and M_{24} also sense the output CM

[10]In addition to occupying a significant area, large resistors also degrade the CM loop stability, as explained later.

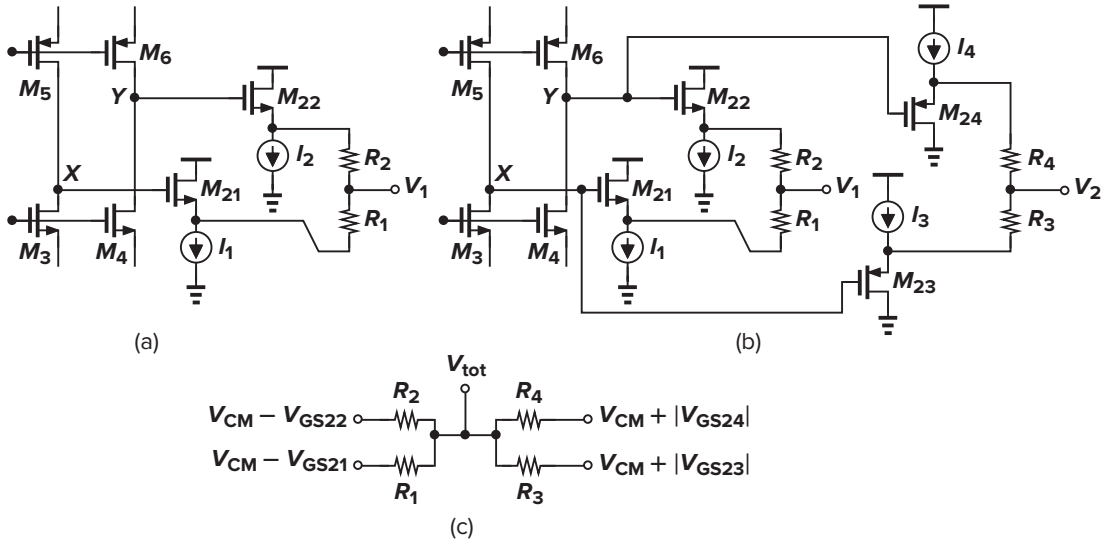


Figure 11.27 (a) CM level reconstruction using NMOS source followers, (b) CM level reconstruction using complementary source followers, and (c) combining network.

level and drive R_3 and R_4, respectively. We recognize that V_1 is lower than V_{CM} by $V_{GS21,22}$ and V_2 is higher than V_{CM} by $|V_{GS23,24}|$:

$$V_1 = V_{CM} - V_{GS21,22} \tag{11.10}$$

$$V_2 = V_{CM} + |V_{GS23,24}| \tag{11.11}$$

We therefore surmise that a linear combination of V_1 and V_2 can remove the V_{GS} terms, yielding a value in proportion to V_{CM}. That is, if

$$\alpha V_1 + \beta V_2 = (\alpha + \beta) V_{CM} - \alpha V_{GS21,22} + \beta |V_{GS23,24}| \tag{11.12}$$

then we must choose $\alpha V_{GS21,22} = \beta |V_{GS23,24}|$, obtaining $\alpha V_1 + \beta V_2 = (\alpha + \beta) V_{CM}$. We also choose $\alpha + \beta = 1$ so that the reconstructed value is equal to the op amp output CM level.

The weighting factors, α and β, can readily be implemented by R_1–R_4 in Fig. 11.27(b). In fact, if V_1 and V_2 are *shorted*, the weighted sum of V_1 and V_2 is produced. With the aid of the equivalent circuit in Fig. 11.27(c), the reader can show that

$$V_{tot} = V_{CM} + \frac{R_N |V_{GS23,24}| - R_P V_{GS21,22}}{R_N + R_P} \tag{11.13}$$

where $R_N = R_1 = R_2$ and $R_P = R_3 = R_4$. We therefore choose $R_N / R_P = V_{GS21,22} / |V_{GS23,24}|$.

In order to evaluate the feasibility of the above idea, we first run a dc sweep in simulations and examine the behavior of V_{tot}. We select a bias current of 10 μA (slightly exceeding the power budget), $W/L = 10\ \mu$m/40 nm for all of the source followers, and $R_N = R_P = 20$ kΩ. The 10-μA bias current sources are also implemented as transistors (with $W/L = 10\ \mu$m/40 nm) to ensure a realistic behavior as V_X and V_Y approach V_{DD} or ground.[11] Figure 11.28 plots the outputs, their actual common-mode level, defined as $(V_X + V_Y)/2$, and the reconstructed counterpart, V_{tot}. We note that V_{tot} closely follows the CM level of V_X and V_Y.

[11] Ideal current sources would allow the source voltages to exceed the supply rails.

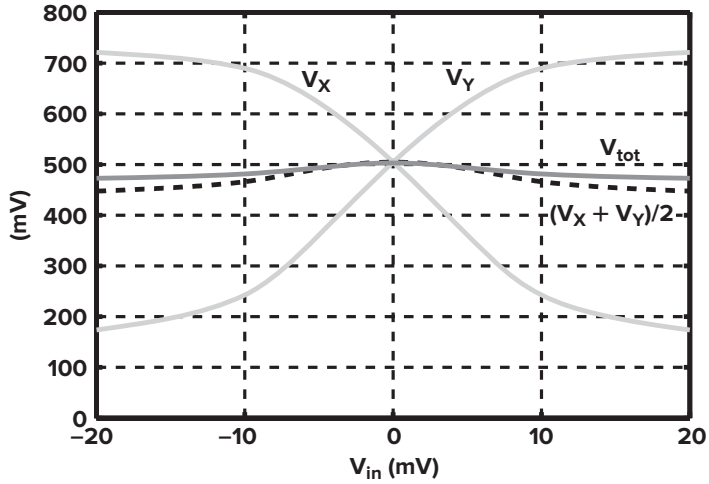


Figure 11.28 Actual CM level, $(V_X + V_Y)/2$, and reconstructed CM level, V_{tot}, of cascode op amp as a function of input differential voltage.

In the next test, let us close the CM feedback loop: we compare V_{tot} to a reference, amplify the error, and return the result to control I_{SS}. To this end, we design the error amplifier as a five-transistor OTA with $W/L = 5\ \mu$m/80 nm for all transistors, a tail current of 20 μA, and a voltage gain of 10. The output of this amplifier controls a fraction of the main tail current, I_1 (Fig. 11.29). For example, if we expect 20% mismatch between the PMOS current sources in the op amp and the tail current source, we choose $I_1 \approx 0.2 I_{SS}$. Figure 11.29 depicts the result, where the OTA's input and output connections are chosen so as to establish *negative* feedback around the loop.

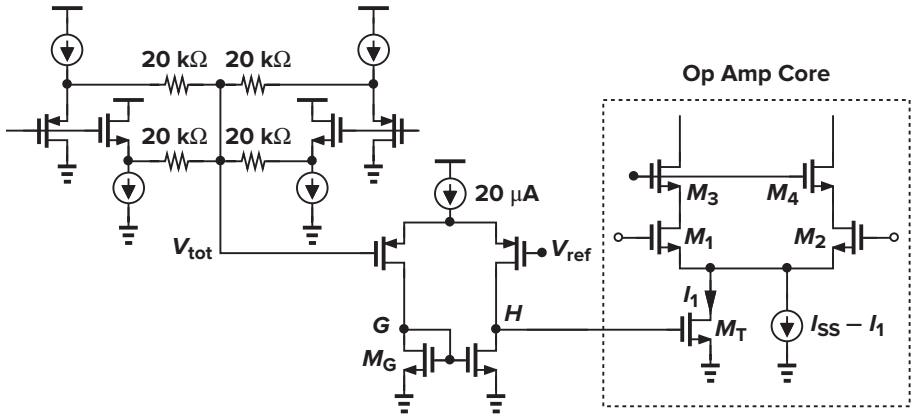


Figure 11.29 CMFB loop around telescopic op amp.

▶ Example 11.7

Explain why the OTA in Fig. 11.29 employs PMOS (rather than NMOS) input devices.

Solution

The choice is dictated by two considerations. First, these transistors must sense the CM level while leaving sufficient V_{DS} for their tail current source. With $V_{tot} \approx V_{DD}/2$ in this case, there is no particular preference for NMOS or PMOS devices. Second, the output of the OTA should have a nominal dc value compatible with that required by M_T.

Since $V_H = V_G$ (in the absence of mismatches), and since V_G is equal to the gate-source voltage of a diode-connected NMOS transistor, we expect that M_T nominally copies the bias current of M_G (with a multiplication factor).

Figure 11.30 shows the closed-loop dc sweep results with $V_{ref} = 0.5$ V.

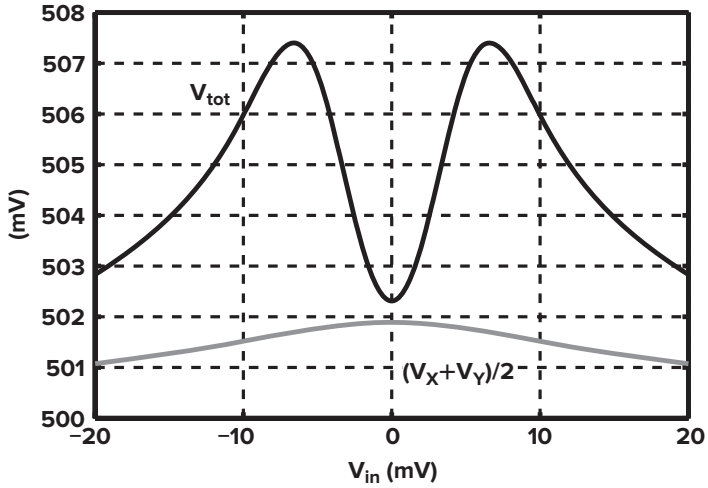


Figure 11.30 Closed-loop behavior of actual CM level, $(V_X + V_Y)/2$, and reconstructed CM level, V_{tot}, of cascode op amp as a function of input differential voltage.

By virtue of feedback, the CM variation is greatly reduced as V_X and V_Y reach high or low values. Next, we create a 10% mismatch between the PMOS current sources (M_7 and M_8 in Fig. 11.21) and $I_{SS}/2 = 950\ \mu$A and repeat the dc sweep. Figure 11.31 depicts the variations, indicating that CMFB suppresses the mismatch by adjusting I_1.

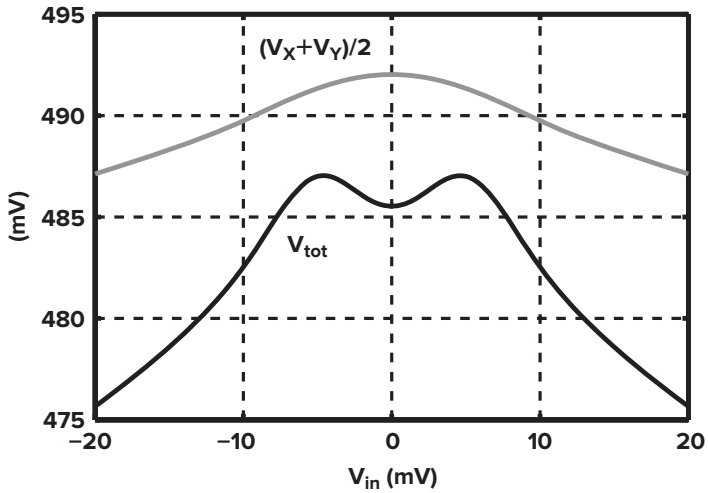


Figure 11.31 Closed-loop behavior of actual CM level, $(V_X + V_Y)/2$, and reconstructed CM level, V_{tot}, of cascode op amp in the presence of mismatch between tail and PMOS current sources.

CMFB Stability We must investigate the stability of the CM loop. This is accomplished by placing the overall op amp in its intended feedback system, applying differential pulses at the input, and examining

the differential and common-mode behavior of the output. Figure 11.32(a) shows a feedback topology for a nominal closed-loop gain of 2, and Fig. 11.32(b) depicts a more detailed diagram highlighting the CM feedback loop.[12]

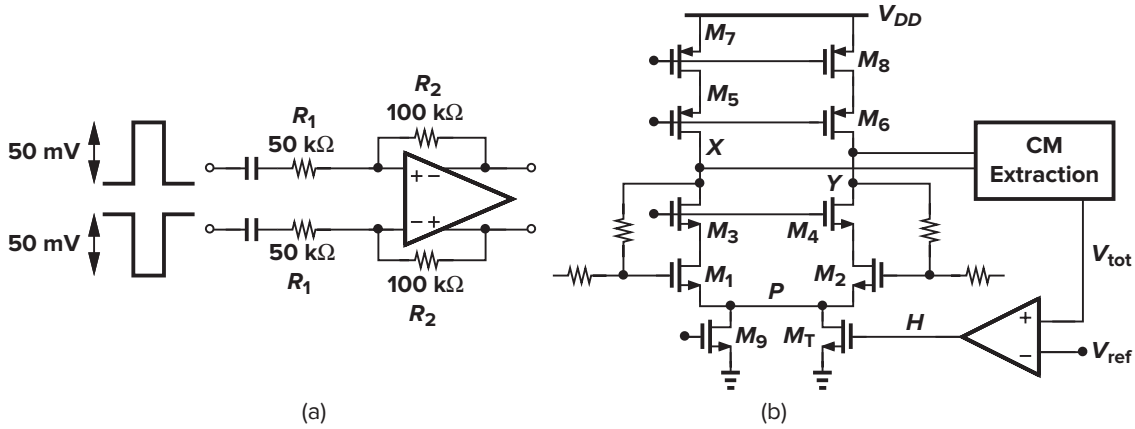


Figure 11.32 (a) Closed-loop amplifier for transient analysis, and (b) detailed view showing the CMFB loop.

Plotted in Fig. 11.33 are the output waveforms in response to an input step, revealing common-mode instability. As evident from Fig. 11.32(b), the CM loop contains a pole at the input of the error amplifier, one at node H, one at node P, one at the sources of the NMOS cascode devices, and one at the main outputs. The loop therefore demands compensation.

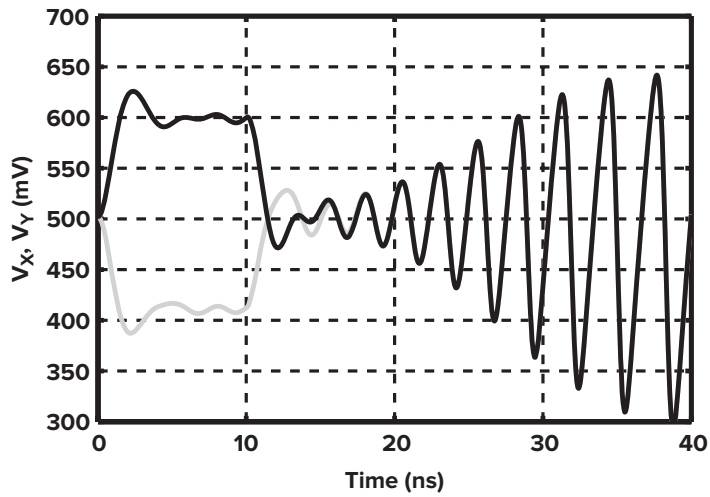


Figure 11.33 Transient response revealing CM loop instability.

▶ Example 11.8

We wish to study the CM loop frequency response and obtain the phase margin. Should the *differential* feedback be present when the CM loop is broken? In other words, which of the two topologies in Fig. 11.34(a) should be used to determine the CM loop transmission?

[12] Since a telescopic-cascode op amp does not easily lend itself to equal input and output CM levels, two 1-μA constant current sources (not shown) are tied from the inputs of the op amp to ground, shifting the input CM level down by 100 mV.

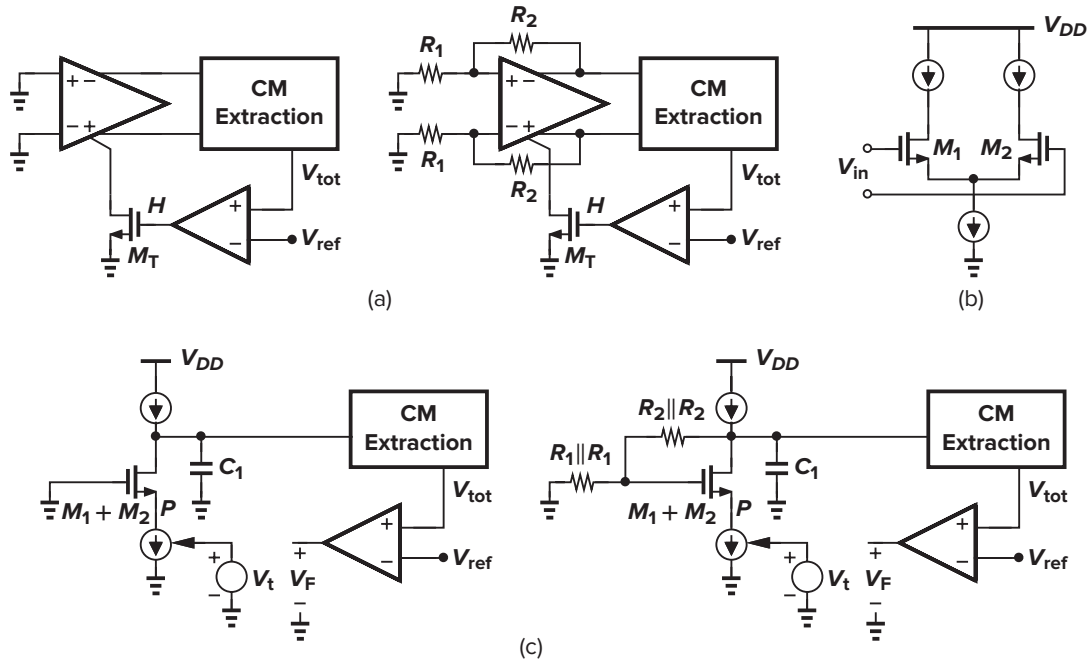


Figure 11.34

Solution

Common-mode feedback must ultimately behave well with differential feedback present. This is because the actual environment in which CMFB must be stable incorporates differential feedback. As an example, consider the simple op amp shown in Fig. 11.34(b). For CM analysis, the two sides can be merged into one, yielding the two possible scenarios depicted in Fig. 11.34(c) if the differential feedback is absent or present. Obtained as $-V_F/V_t$, the CM loop transmissions derived for these two cases are not necessarily the same. For example, if the capacitance at the drain, C_1, is considered, the pole associated with this node assumes different values in the two topologies. Thus, we must maintain differential feedback while studying CM stability.

◀

Let us break the CM loop in Fig. 11.32(b) at node H as shown in Fig. 11.35. Here, the error amplifier drives a dummy device, M_d, identical to M_T so as to see the loading effect of the latter. Plotted in

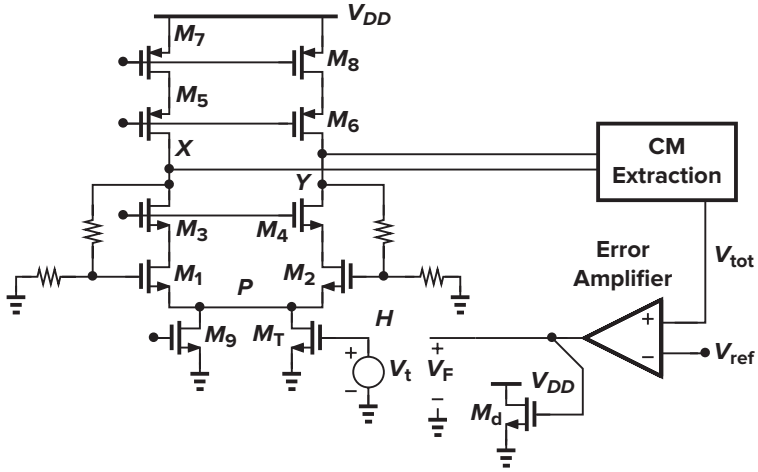


Figure 11.35 Measurement of CMFB loop transmission.

Fig. 11.36(a) are the magnitude and phase of $-V_F/V_t$ as a function of frequency, revealing a phase of -190° at the unity-gain frequency. We seek a convenient node for compensation. Unfortunately, the error amplifier in Fig. 11.29 does not provide signal inversion from V_{tot} to H and hence cannot employ Miller compensation.

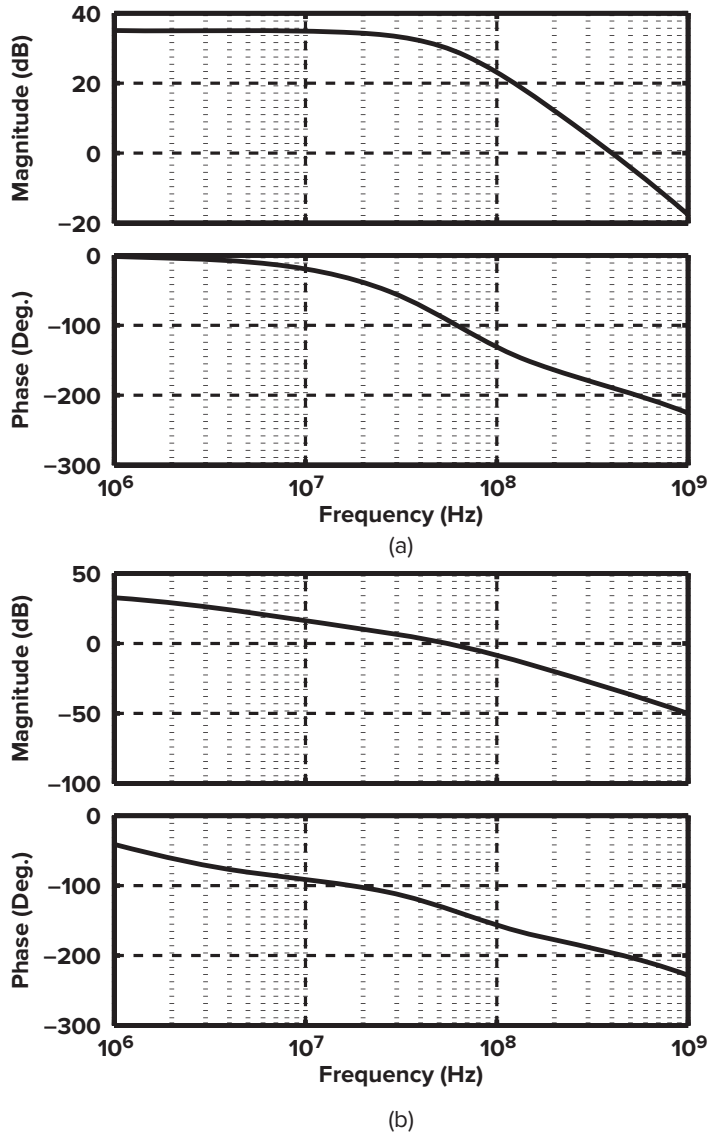


Figure 11.36 CMFB loop transmission (a) before and (b) after compensation.

Can we compensate the CM loop by adding capacitance from high-impedance nodes X and Y to ground? Yes, but this also affects the differential response. Instead, we tie a 3-pF capacitor from the error amplifier output to ground, obtaining the response shown in Fig. 11.36(b) and a phase margin of about 50°. The closed-loop pulse response depicted in Fig. 11.37(a) implies that the common-mode feedback loop is now properly compensated and the CM level incurs little ringing.

Differential Compensation Why do V_X and V_Y in Fig. 11.37(a) exhibit *differential* ringing? This is because the pole formed by the large resistors in the feedback network of Fig. 11.32(a) and the input

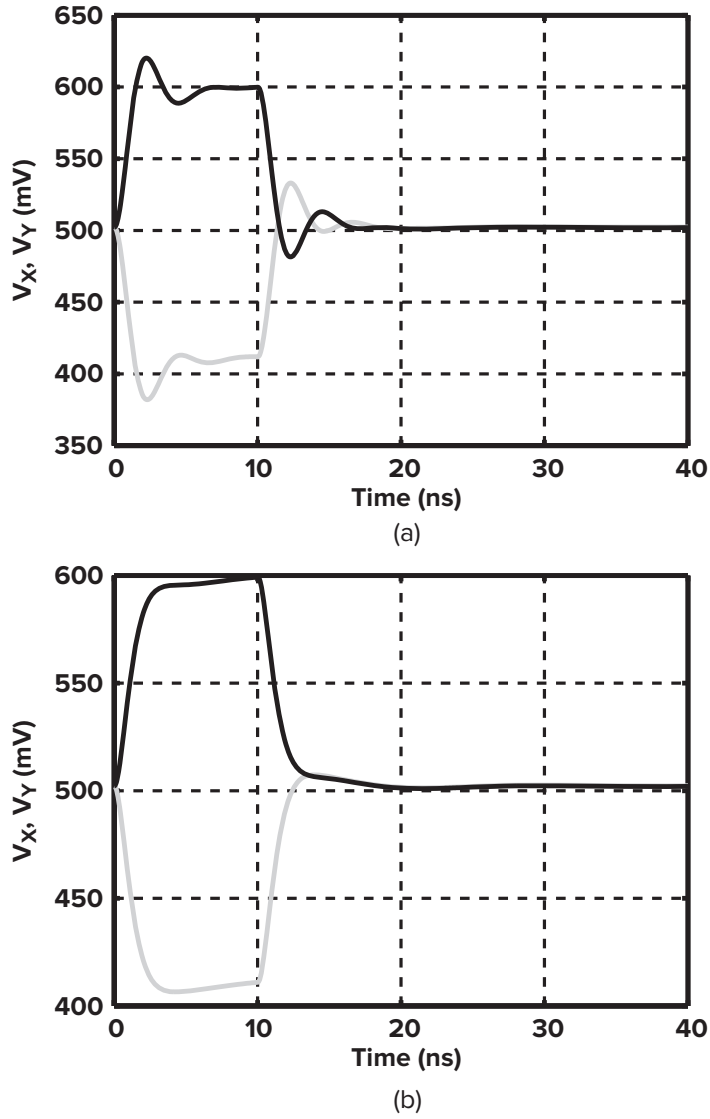


Figure 11.37 Transient response with (a) CMFB loop compensation and (b) additional differential compensation.

capacitance of the op amp is located at a low frequency, degrading the phase margin (of differential feedback). To compensate the differential signal path, we connect two 7-fF capacitors from the outputs of the op amp to its inputs (in parallel with the feedback resistors) so as to create Miller multiplication. Shown in Fig. 11.37(b), the resulting response is now well behaved. This pole-zero cancellation technique is studied in Problem 11.14.

▶ Example 11.9

Explain what design modifications are necessary if the op amp drives a significant load capacitance, C_L.

Solution

The load capacitance lowers the magnitude of the pole at X (and Y), increasing the differential signal path phase margin (Chapter 10) while decreasing the CM loop phase margin. For this reason, the capacitance tied to the error

amplifier output must be increased, or the pole at X and Y must become the dominant pole for the CM loop as well.

◀

Design Summary In this section, we have attempted to design a telescopic-cascode op amp for a voltage gain of 500 and a differential output swing of 1 V_{pp}. Neither specification could be met with a 1-V supply, but we have established the steps that one must complete in order to arrive at the final design. Specifically, we have dealt with the following general principles:

1. Allocation of V_{DS} and I_D to transistors according to required swings and power dissipation, respectively
2. Characterization and scaling of MOSFETs for allowable V_{DS} and desired current level
3. Quick estimate of the achievable voltage gain
4. Use of dc sweep to study bias conditions and nonlinearity
5. Design of bias circuitry using current mirrors and low-voltage cascodes
6. Common-mode feedback design and compensation
7. Use of closed-loop transient analysis to study CM and differential stability

As seen in subsequent sections, these principles provide a systematic approach to the design of op amps.

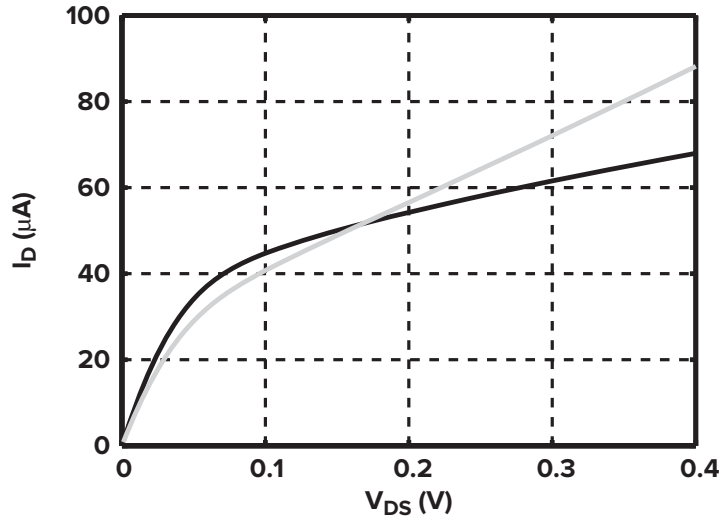
The next natural candidate for our op amp design is the folded cascade. However, our gain calculations for the telescopic cascode roughly apply here as well, predicting that it is extremely difficult to achieve a gain of 500. For this reason, we do not pursue the folded-cascode topology for these specifications.

11.5.2 Two-Stage Op Amp

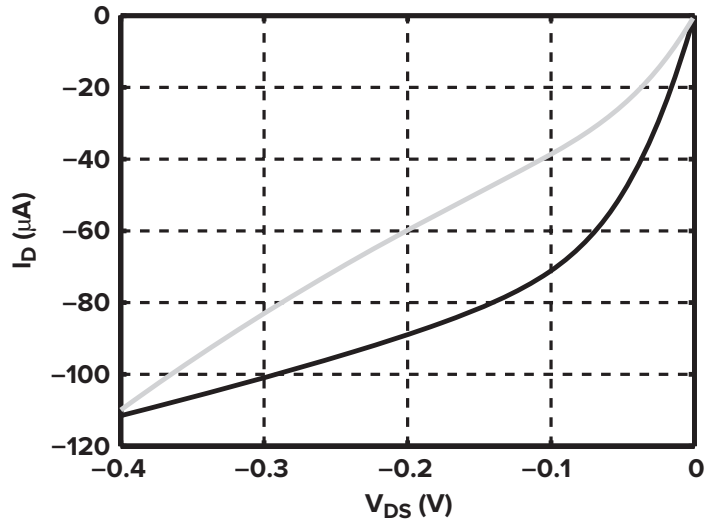
Both the relatively high voltage gain and the 1-V_{pp} swing point to a two-stage configuration as a feasible candidate. We note that a gain of 500 dictates the use of cascoding in the first stage, encouraging us to utilize the telescopic design from the previous section (Fig. 11.21). However, two points must be borne in mind. First, the previous design exhausts the power budget, leaving none for the second stage. Second, with a gain of about 50 in the first stage, the gain of the second stage can be around 10. Thus, the single-ended peak-to-peak swing at the outputs of the first stage can be as small as 50 mV, allowing us to redesign the cascode for greater V_{DS}'s and hence more robust operation.

We must first partition the power budget between the two stages, a task requiring speed and/or noise specifications. We split the power equally here; further optimization could be pursued after one round of complete design. With about 100 μA reserved for the bias network, we allocate 1.9 mA/4 = 475 μA to each branch of transistors in the first and second stages.

First-Stage Design The telescopic-cascode configuration must accommodate a single-ended swing of 50 mV_{pp}, allowing 0.95 V for the sum of five V_{DS}'s. With some margin, we choose $V_{DS,N} = 150$ mV and $V_{DS,P} = 200$ mV and simulate our reference transistors ($W/L = 5\ \mu$m/40 nm and 10 μm/80 nm), seeking acceptable knee voltages. Shown in Fig. 11.38 for $V_{GS,N} = 350$ mV and $V_{GS,P} = -450$ mV, the characteristics exhibit substantially higher drain currents than those in Sec. 11.5.1. It is interesting to note that, as a result of velocity saturation, the knee voltage has not increased by 50 mV. The width of the NMOS transistors in the signal path must be scaled by a factor of 450 μA/50 μA for either $L = 40$ nm or $L = 80$ nm. Similarly, the width of the PMOS device must be scaled by a factor of 450 μA/90 μA for $L = 80$ nm. We also choose for the tail current device $W = (900\ \mu\text{A}/50\ \mu\text{A}) \times 10\ \mu$m and $L = 80$ nm. The cascode-stage devices are thus much narrower than those used in the previous section. The first-stage design is shown in Fig. 11.39(a) and its simulated behavior in Fig. 11.39(b), revealing a gain of about 50. The biasing of this stage is similar to that described in Sec. 11.5.1.



(a)



(b)

Figure 11.38 I_D-V_{DS} characteristics for (a) an NMOS device with $V_{GS} = 350$ mV and $W/L = 5\ \mu$m/40 nm (gray plot) or 10 μm/80 nm (black plot), and (b) a PMOS device with $V_{GS} = -450$ mV and $W/L = 5\ \mu$m/40 nm (gray plot) or 10 μm/80 nm (black plot).

▶ Example 11.10

In order to determine, with the aid of simulation, the small-signal resistance seen at node X in Fig. 11.39(a), a student sets the input signal to zero, ties a unit ac current source from this node to ground, and measures the resulting voltage. Explain why this test overestimates the resistance.

Solution

The voltage developed at X causes a current to flow through r_{O3} to the drain of M_1 and through r_{O1} to the source of M_2. In other words, since M_1 is degenerated by M_2, the resistance at X is overestimated. To avoid this error, a large

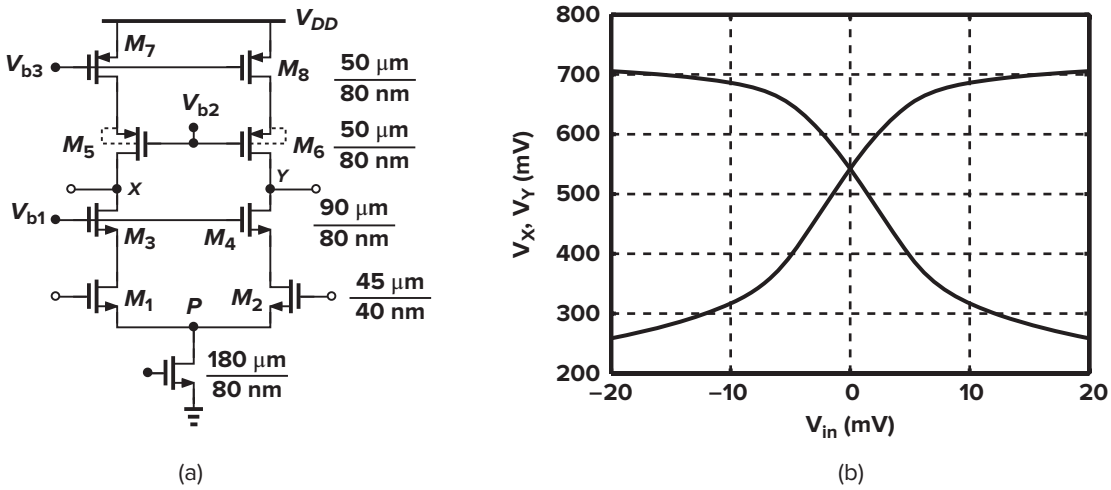


Figure 11.39 (a) First stage design, and (b) its input-output characteristics.

capacitance must be tied from the source of M_1 to ground to create a short circuit at the test frequency. Alternatively, we can attach the ac current source between X and Y, measure the resistance, and divide the result by two.

◀

Second-Stage Design The second stage must provide a voltage gain of about 10, dictating channel lengths greater than 40 nm for both NMOS and PMOS devices. Do we use an NFET or a PFET for the input of the second stage? The need for gain may point to an NFET due to its higher $g_m r_O$, but we must examine the situation more closely. Bearing in mind that the output CM level of the first stage is around 0.55 V, let us consider a transistor having $W/L = 10\ \mu$m/80 nm and determine its $g_m r_O$ if it is an NFET with $V_{GS} \approx 0.55$ V or a PFET with $|V_{GS}| \approx 0.45$ V. Using simulations, we obtain $(g_m r_O)_N = 12.8$ and $r_{ON} = 1.86\ \text{k}\Omega$ at $V_{DS} = 0.5$ V and $I_D = 900\ \mu$A, and $(g_m r_O)_P = 17.5$ and $r_{OP} = 9.75\ \text{k}\Omega$ at $|V_{DS}| = 0.5$ V and $|I_D| = 110\ \mu$A. We therefore select the PFET and scale its width to $(450\ \mu\text{A}/110\ \mu\text{A}) \times 10\ \mu\text{m} \approx 41\ \mu$m to accommodate the nominal bias current. With $W/L = 41\ \mu$m/80 nm, such a device exhibits an output resistance of 2.38 kΩ. The drain of the PFET is tied to an NMOS current source.

The NMOS current source output resistance must not lower the gain of the second stage, $|A_{v2}|$, below 10. Writing $|A_{v2}| = g_{mP}(r_{OP}||r_{ON}) \geq 10$, we have $r_{ON} \geq 1.33 r_{OP} = 3.0\ \text{k}\Omega$ at $I_D = 475\ \mu$A. If the 10-μm/80-nm NFET considered above with $r_O = 1.86\ \text{k}\Omega$ and $I_D = 900\ \mu$A is scaled down by a factor of 2, it yields $r_{ON} = 3.72\ \text{k}\Omega$, which is close to the desired value.

Figure 11.40(a) shows the op amp developed thus far, and Fig. 11.40(b) plots the input-output characteristics. In order to determine the maximum output swing that the op amp can handle, we plot the slope of the differential characteristic in Fig. 11.40(c), noting that the differential output cannot exceed 450 mV if the gain must not drop below 500. To resolve this issue, we double the width and length of the output NFETs, raising the gain and arriving at the results depicted in Fig. 11.41. Now, the single-ended swing reaches 530 mV for a minimum gain of 500. Of course, the gain *variation* (nonlinearity) is unabated, posing difficulties in some applications.

Common-Mode Feedback As explained in Chapter 9, two-stage op amps generally require CMFB for both stages. For the first stage, we can utilize the CMFB scheme illustrated in Fig. 11.29. We therefore focus on CMFB for the second stage.

The second stage can also incorporate the method of Fig. 11.29 and control the NMOS current sources. However, the lower output impedance here allows the use of resistors for direct sensing of the CM level,

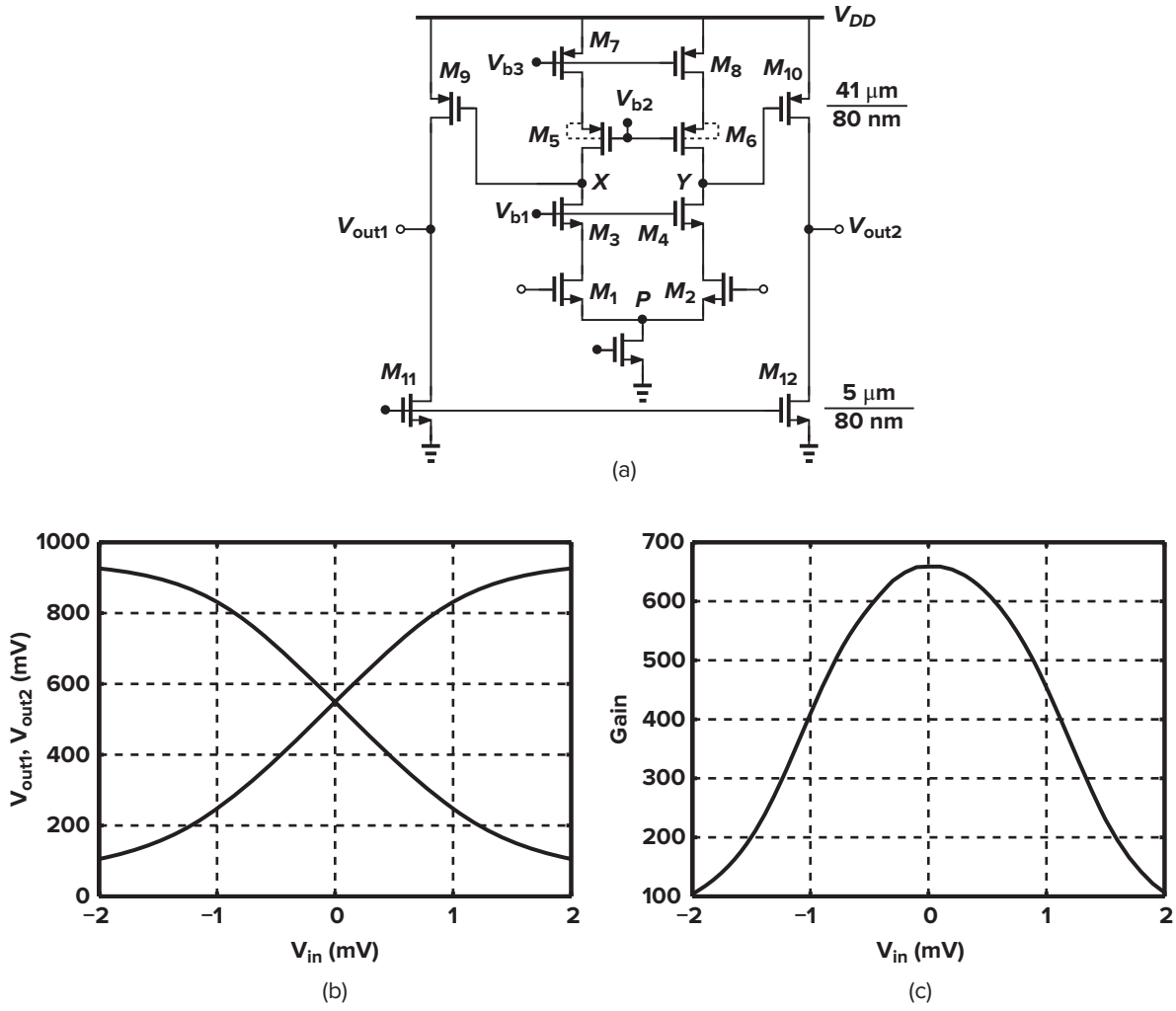


Figure 11.40 (a) Two-stage op amp design, (b) its input-output characteristics, and (c) its gain variation.

simplifying the design. Consider the topology depicted in Fig. 11.42(a), where R_1 and R_2 ($\approx$ 30 kΩ) reconstruct the CM level at node G, applying the result to the gates of M_{11} and M_{12}. Under equilibrium, the resistors draw no current, establishing an output CM voltage equal to $V_{GS11,12}$. This voltage varies by about 50 mV with PVT, a value that can be tolerated in this design. Note that this CMFB loop is stable.

What if $V_{GS11,12}$ is not close to the desired output CM level? As shown in Fig. 11.42(b), if we inject a current I_B into node G, the output CM level is shifted by $I_B R_1/2 (= I_B R_2/2)$. For example, a shift of 100 mV requires a current of (100 mV/30 kΩ) $\times$ 2 = 6.7 μA. A positive I_B shifts the CM level down and vice versa.

Frequency Compensation The two-stage op amp designed above contains several poles and most likely demands compensation. Recall from Chapter 10 that the first nondominant pole of two-stage op amps typically arises from the output node and hence depends on the load capacitance, C_L. The stability

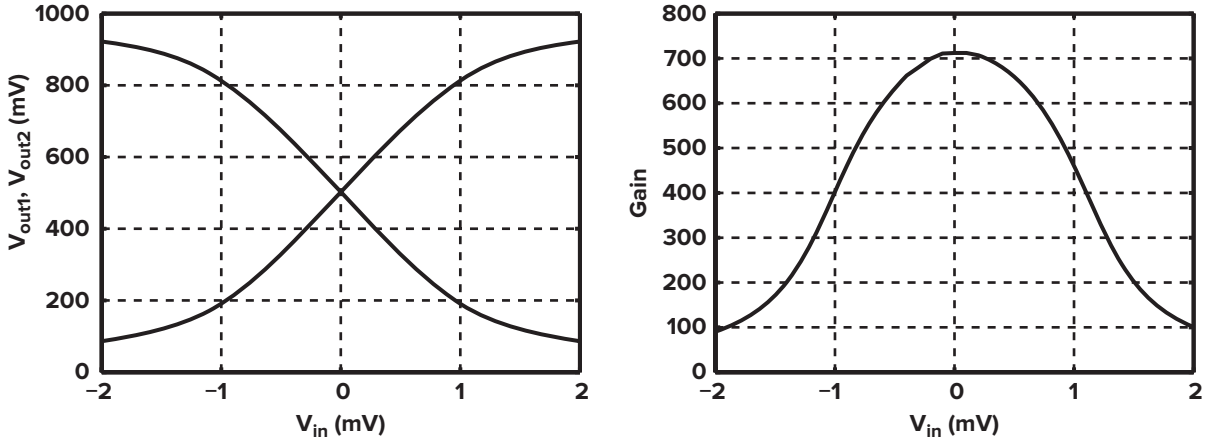


Figure 11.41 (a) Input-output characteristics and (b) gain variation of modified two-stage op amp with $(W/L)_{11,12} = 10\ \mu$m/0.16 μm.

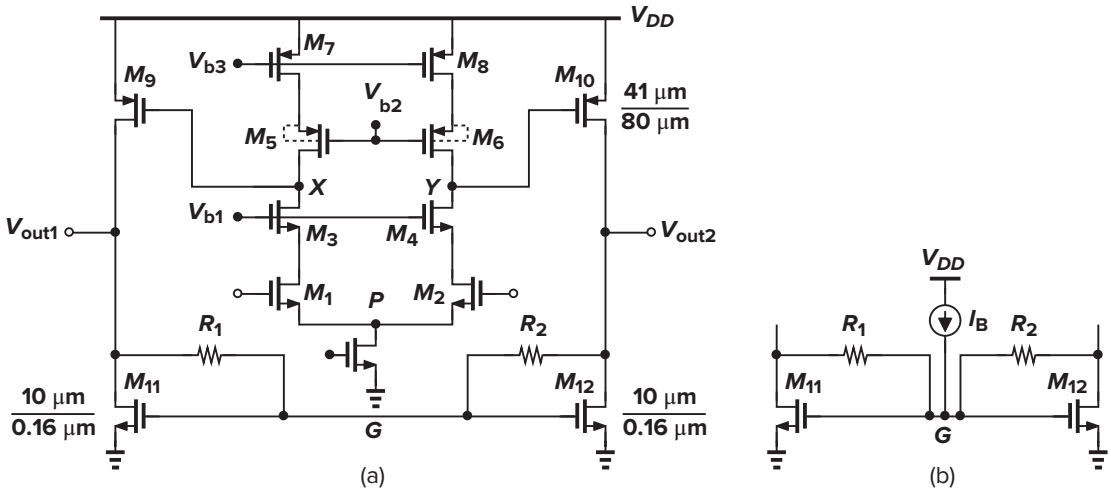


Figure 11.42 (a) Simple common-mode feedback around the second stage; (b) injection of current to shift CM level.

analysis must therefore assume a value of C_L, which itself is given by the environment in which the op amp is used. Let us choose a single-ended load capacitance of 1 pF in this example, obtaining an output pole frequency of around 90 MHz. We begin the study with the open-loop op amp, bearing in mind that the feedback network may add its own effects and eventually require further modifications.

Plotted in Fig. 11.43 are the open-loop (differential) magnitude and phase response of the circuit, revealing a low-frequency gain of 57 dB (≈ 700), a unity-gain frequency of 3.2 GHz, and a phase margin of about -8°. This bandwidth appears very impressive, but we also note that the phase reaches -120° at 240 MHz. In other words, after compensation for 60° phase margin, the unity-gain bandwidth can drop by a factor of 13!

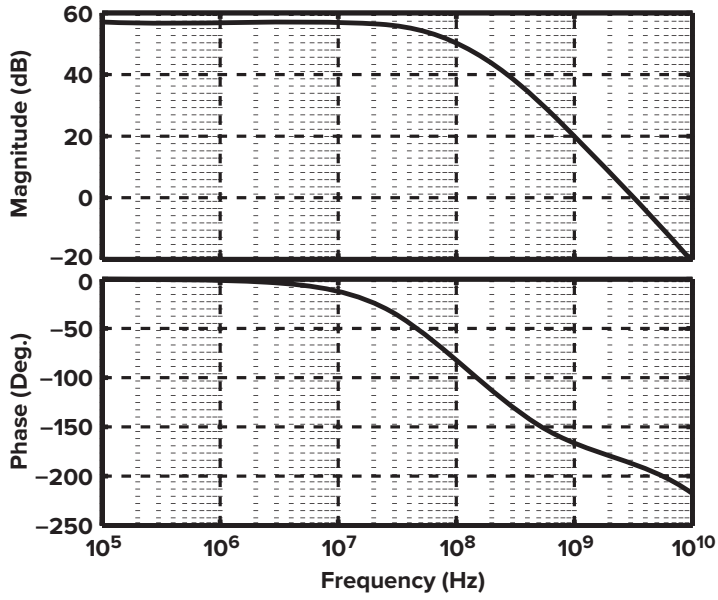


Figure 11.43 Frequency response of open-loop op amp.

▶ **Example 11.11**

The above results are rather curious: the output pole is located around 90 MHz, suggesting a phase of about -135° at this frequency, but the actual phase is around -85° at this frequency. Explain the reason for this behavior.

Solution

In the above design, we cannot say that the phase reaches -135° at the second pole because the poles are not widely spaced. In fact, the pole at X in Fig. 11.42 is around 95 MHz. The pole at X and the output pole produce at 90 MHz a phase shift of $-\tan^{-1}(90/95) - \tan^{-1}(90/90) \approx -88^\circ$.

We express the phase shift at 240 MHz due to these two poles as $-\tan^{-1}(240\text{ MHz}/95\text{ MHz}) - \tan^{-1}(240\text{ MHz}/90\text{ MHz}) = -138^\circ$. Why does this result disagree with the simulated value of -120°? This is because the gate-drain capacitance of the output PMOS transistor creates some pole splitting, raising the output pole beyond 90 MHz and lowering the pole at X below 95 MHz. ◀

In order to compensate the op amp, we begin at 240 MHz and 0 dB on the magnitude plot in Fig. 11.43 and draw a straight line toward the y-axis with a slope of -20 dB/dec. The frequency at which this line intersects the magnitude plot is roughly equal to 240 MHz/700 = 344 kHz (why?), yielding the desired value for the dominant pole.

Which node should produce the dominant pole: X or the output node? We prefer the former for two reasons, namely, a smaller compensation capacitance due to Miller multiplication and pole splitting; neither of these benefits accrues if the dominant pole is established at the output.

With the output resistance of 8 kΩ seen at node X and a voltage gain of about 12 provided by the output stage, we choose a Miller compensation capacitor, C_C, equal to 4.5 pF so as to create a 344-kHz pole at this node. Figure 11.44 shows the resulting open-loop frequency response, confirming that the dominant pole is now located around 340 kHz. Unfortunately, the gain crossover occurs at 350 MHz and the phase margin is only 18° because the zero introduced by C_C, $\omega_z = g_{m10}/C_C$, is as low as 250 MHz. As explained in Chapter 10, we can insert a resistor, R_z, in series with C_C so as to move the zero to the second pole, ω_{p2}. The second pole can be roughly estimated from Fig. 11.44 as the frequency at

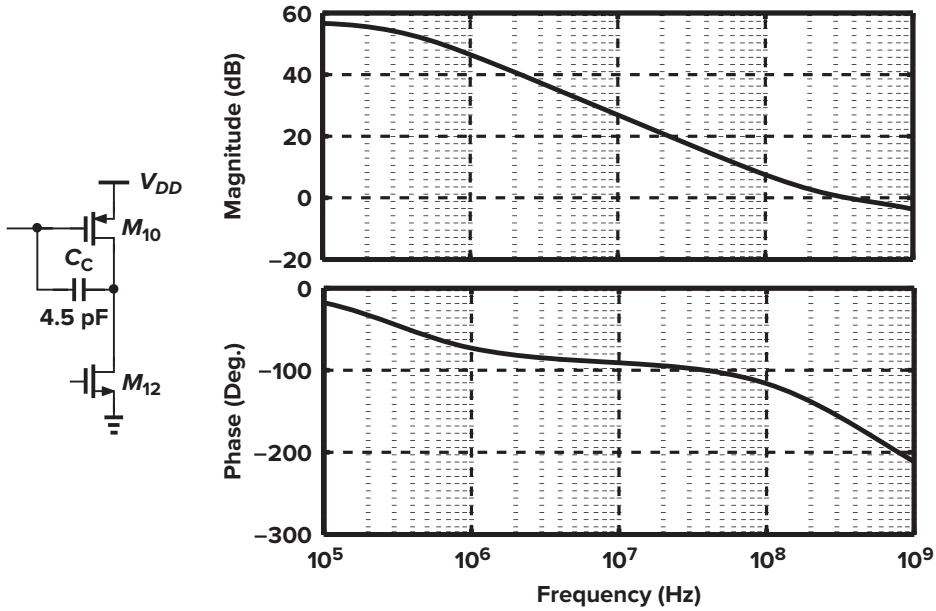


Figure 11.44 Frequency response of compensated open-loop op amp with $C_C = 4.5$ pF.

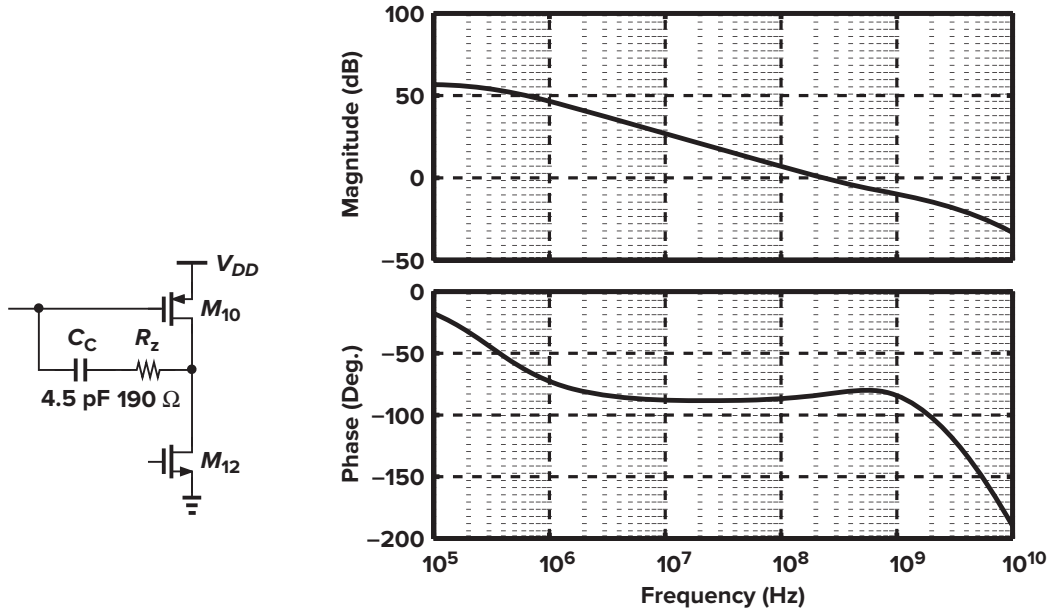
which $\angle H$ reaches -135° and is equal to 185 MHz. Selecting R_z according to $(\omega_{p2}C_C)^{-1} = 190\ \Omega$, we observe the response depicted in Fig. 11.45(a). The phase margin rises to 96° because of the pole-zero cancellation.

The phase margin revealed by Fig. 11.45(a) suggests that the compensation capacitor can be smaller and the unity-gain bandwidth larger. By some iteration, we choose $C_C = 0.8$ pF and $R_z = 450\ \Omega$, arriving at the response shown in Fig. 11.45(b). Remarkably, the op amp now exhibits a unity-gain bandwidth of 1.9 GHz with a phase margin of 65°.

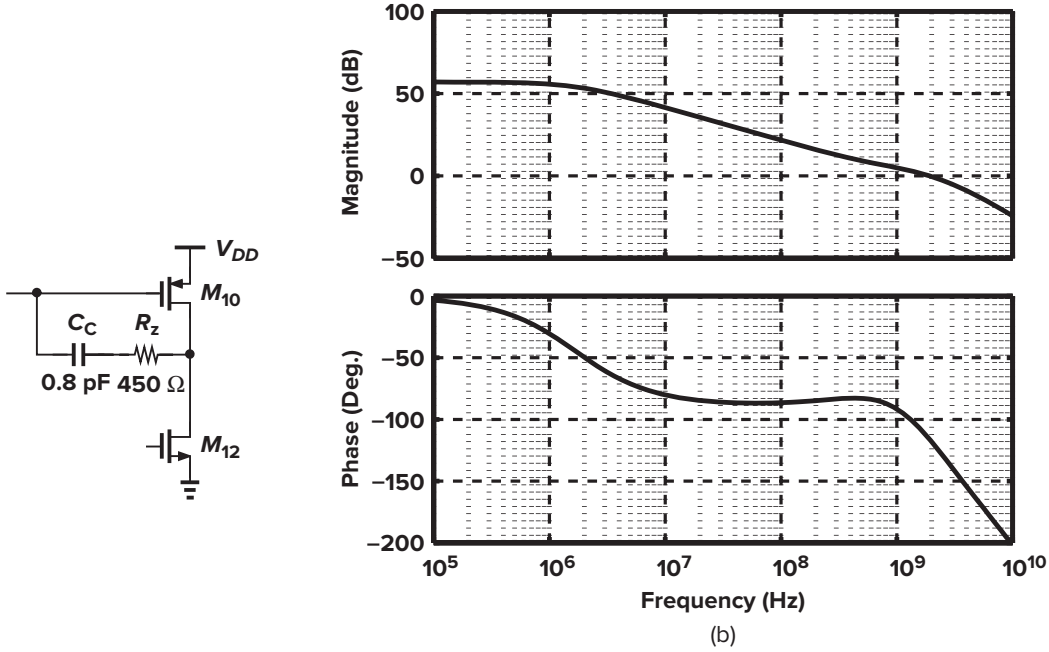
Closed-Loop Behavior We now configure the op amp as a closed-loop amplifier having a nominal gain of 2 and a load capacitance of 1 pF [Fig. 11.46(a)]. The small-signal transient response appears as shown in Fig. 11.46(b), exhibiting significant ringing. Why does this happen despite the 65° phase margin obtained above? This is due to the large resistance values used in the feedback network. We draw the single-ended equivalent as in Fig. 11.47(a) for the loop transmission calculation, observing that an open-loop pole around $[2\pi(100\ \text{k}\Omega||50\ \text{k}\Omega)C_{in}]^{-1} \approx 95$ MHz is formed at the input of the op amp.

In order to improve the closed-loop stability, we can reduce R_1 and R_2 in Fig. 11.47(a) to, say, 25 kΩ and 50 kΩ, respectively, before the open-loop gain falls appreciably, but this remedy only doubles the input pole frequency. Alternatively, we can increase the resistance in series with the compensation capacitors from 450 Ω to 1500 Ω, arriving at the response shown in Fig. 11.47(b). The circuit now settles much faster.

We conclude this section with two remarks. First, the op amp has been compensated for unity-gain feedback, whereas the topology of Fig. 11.46 operates with a feedback factor of $50\ \text{k}\Omega/150\ \text{k}\Omega = 1/3$. Thus, the compensation capacitance can be reduced to lower the phase margin to around 60°. Second, the design has assumed the "typical-NMOS, typical-PMOS" (TT) corner of the process, a temperature of 27°C, and a constant supply of 1 V. In practice, we must account for other corners (e.g., SS or FF), the required temperature range (e.g., 0°C to 75°C), and supply variations (e.g., by ±5%). To meet the specifications under all of these conditions, the design must often be more conservative in terms of gain, swings, and power consumption than that presented here.



(a)



(b)

Figure 11.45 Frequency response of open-loop amp with (a) $C_C = 4.5$ pF and $R_Z = 190\ \Omega$, and (b) $C_C = 0.8$ pF and $R_Z = 450\ \Omega$.

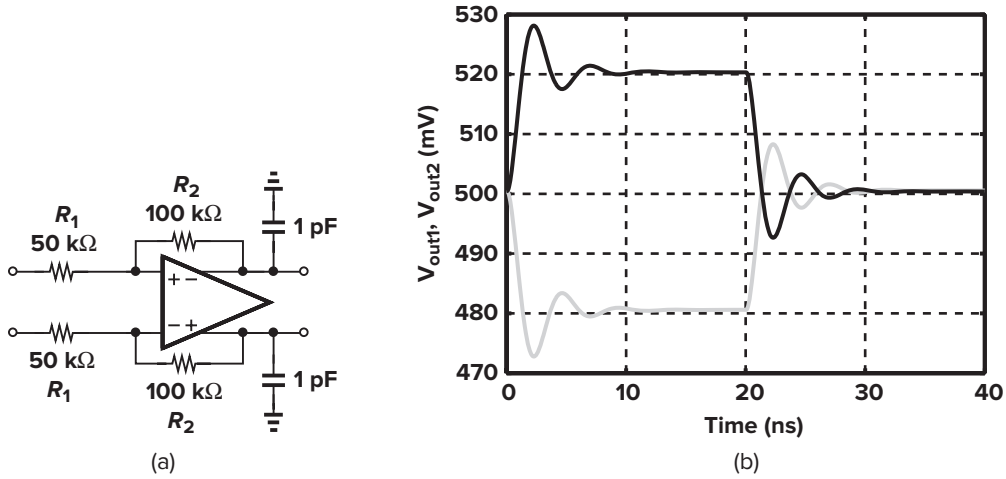


Figure 11.46 (a) Closed-loop amplifier and (b) its step response.

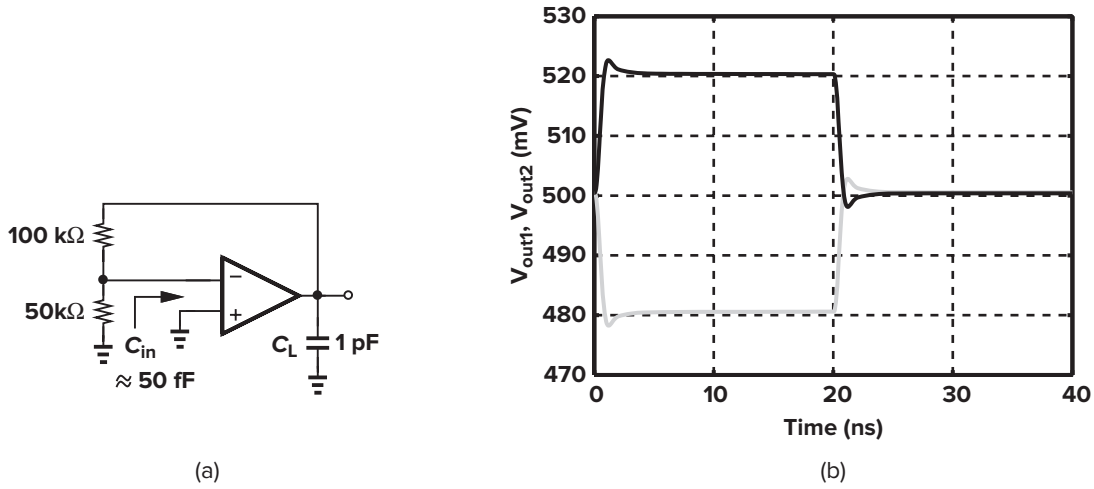


Figure 11.47 (a) Equivalent circuit of closed-loop amplifier and (b) step response with $R_z = 1500\ \Omega$.

11.6 ■ High-Speed Amplifier

Some applications require an amplifier with fast settling and accurate gain. For example, "pipelined" ADC architectures can tolerate but a small gain error in their constituent amplifiers. In this section, we design a differential amplifier according to the following specifications:

- Voltage gain = 4
- Gain error $\leq 1\%$
- Differential output swing = 1 V_{pp}
- Load capacitance = 1 pF
- Step response settling time to 0.5% accuracy = 5 nS
- $V_{DD} = 1$ V

As illustrated in Fig. 11.48, the settling time, t_s, is defined as the time necessary for the output to reach within 0.5% of its final value. Our objective is to minimize the power consumption of the circuit.

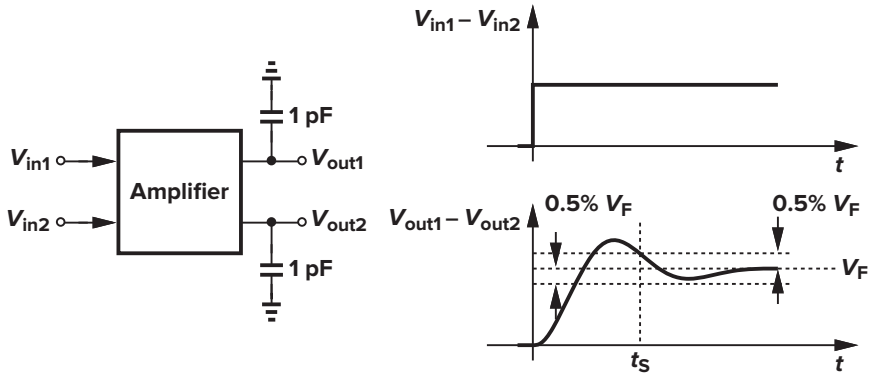


Figure 11.48 Definition of settling time.

11.6.1 General Considerations

Precision Issues With a myriad of amplifier topologies, where do we begin? In this case, the specifications readily narrow down our choices. The maximum tolerable gain error of 1% indicates a *closed-loop* configuration so that the gain can be defined as the ratio of two passive component values and remain relatively independent of PVT. We must therefore design an amplifier whose open-loop gain is high enough to yield a closed-loop gain error of less than 1%. This observation along with the required output swing of 1 V_{pp} calls for a two-stage op amp.

We have now arrived at the feedback arrangement shown in Fig. 11.49, where the closed-loop gain is given by

$$\frac{V_{out}}{V_{in}} = -\frac{R_2}{R_1}\frac{1}{1+(1+\dfrac{R_2}{R_1})\dfrac{1}{A_0}} \tag{11.14}$$

$$\approx -\frac{R_2}{R_1}\left[1-\left(1+\frac{R_2}{R_1}\right)\frac{1}{A_0}\right] \tag{11.15}$$

We choose $R_2/R_1 = 4$ and ensure that the gain error falls below 1%:

$$\left(1+\frac{R_2}{R_1}\right)\frac{1}{A_0} \le 0.01 \tag{11.16}$$

thereby obtaining $A_0 \ge 500$. This calculation neglects the loading of the feedback network on the op amp.

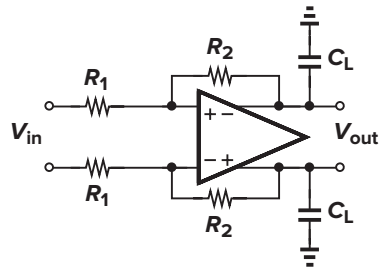


Figure 11.49 Closed-loop amplifier with resistive feedback.

▶ Example 11.12

Determine the closed-loop output impedance and bandwidth of the above topology in terms of the open-loop op amp characteristics.

Solution

Drawing the half-circuit equivalent for loop gain calculation as shown in Fig. 11.50 and applying a test signal, V_t, we observe that the feedback network senses V_{out} and returns a fraction equal to $\beta = [R_1/(R_1 + R_2)]V_{out}$ to the input. The loop gain is therefore equal to $\beta A_0 = A_0 R_1/(R_1 + R_2) \approx A_0/5 \approx 100$, indicating that the output resistance falls by a factor of 100 as a result of feedback. The bandwidth rises by the same factor.

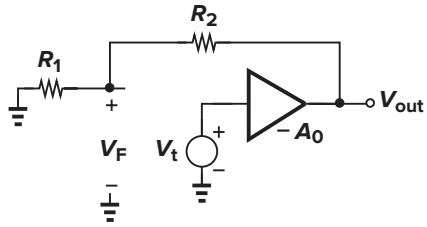


Figure 11.50

◀

The use of a resistive feedback network poses a difficulty: as explained in Sec. 11.5.2, if R_1 and R_2 are large enough not to reduce the open-loop gain of the op amp, then they form a significant pole with the input capacitance, degrading the phase margin. We therefore consider *capacitive* feedback instead, configuring the circuit as shown in Fig. 11.51(a). The closed-loop gain is now approximately equal to C_1/C_2, or more accurately (Chapter 13):

$$\frac{V_{out}}{V_{in}} \approx -\frac{C_1}{C_2}\left(1 - \frac{C_1 + C_2 + C_{in}}{C_2}\frac{1}{A_0}\right) \tag{11.17}$$

where C_{in} denotes the (single-ended) input capacitance of the op amp. Drawing the single-ended counterpart for the loop transmission calculation [Fig. 11.51(b)], we observe that C_1 and C_2 do *not* contribute additional poles because $(C_1+C_{in})C_2/(C_1+C_{in}+C_2)$ simply appears in parallel with C_L. (As explained in Chapter 13, the capacitors also allow sampling and discrete-time operation.)

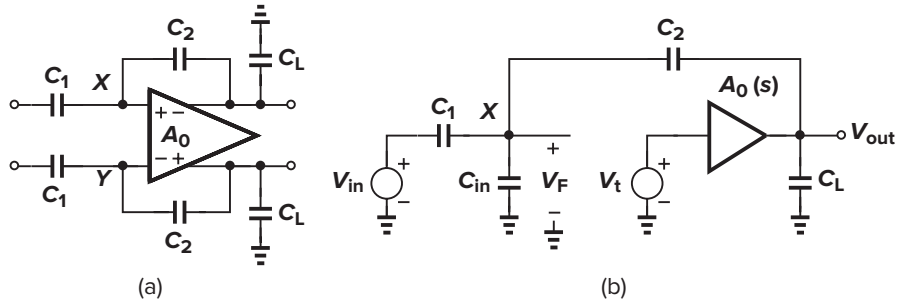


Figure 11.51 (a) Closed-loop amplifier using capacitive feedback, and (b) its simplified equivalent for loop gain calculation.

▶ Example 11.13

Circuits incorporating capacitive "coupling" typically exhibit a high-pass response. Is that the case for the amplifier of Fig. 11.51(a)?

Solution

No, it is not. Since there is no resistive path to X and Y, the time constant at these nodes is infinite (if leakage currents are neglected), yielding a frequency response that extends to nearly $f = 0$. The reader can consider the frequency response of a simple capacitive divider as an example to appreciate this property. ◀

The circuit of Fig. 11.51(a) provides no bias for the op amp inputs, i.e., the dc levels at X and Y are not defined and can assume any value. (In the presence of gate leakage at the inputs, these nodes charge to V_{DD} or discharge to ground.) Illustrated in Fig. 11.52, a simple remedy is to add two feedback resistors so that the input and output dc levels become equal. However, the finite time constant at X and Y leads to a high-pass response; if $A_0 = \infty$, then

$$\frac{V_{out}}{V_{in}}(s) = -\frac{R_F \| \dfrac{1}{C_2 s}}{\dfrac{1}{C_1 s}} \tag{11.18}$$

$$= -\frac{R_F C_1 s}{R_F C_2 s + 1} \tag{11.19}$$

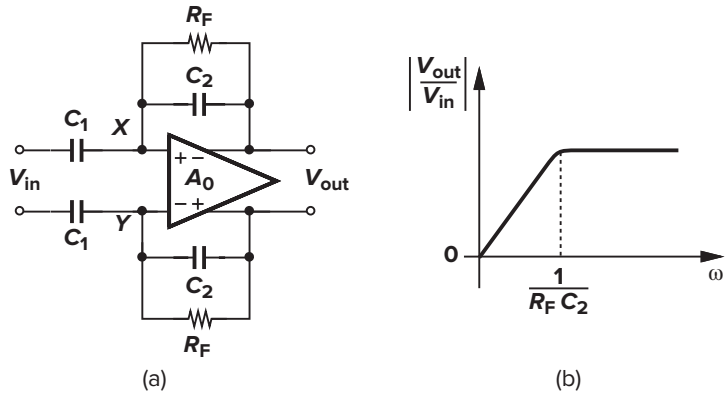


Figure 11.52 Addition of feedback resistors to define input dc levels and the resulting transfer function.

The corner frequency, $1/(2\pi R_F C_2)$, must therefore be chosen less than the minimum input frequency of interest, a condition that is not practical in all applications. As explained in Chapter 13, R_F can be replaced with a switch, but we proceed here assuming that $R_F C_2$ is sufficiently large. In other words, we assume that the circuit reduces to that in Fig. 11.51(a) for the frequencies of interest.

Equation (11.17) indicates that the capacitive-feedback amplifier's gain error also depends on C_{in}. For example, if $C_{in} \approx (C_1 + C_2)/5$, then A_0 must be 20% higher than the value dictated by Eq. (11.16). We can choose $C_1 + C_2 \gg C_{in}$, but at the cost of settling speed.

Speed Issues The amplifier must settle to 0.5% accuracy in 5 ns. Let us first assume a linear, first-order circuit and write the step response as

$$V_{out}(t) = V_0 \left(1 - \exp\frac{-t}{\tau}\right) u(t) \tag{11.20}$$

The time necessary for V_{out} to reach $0.995V_0$ is $t_s = -\tau \ln 0.005 = 5.3\tau$; i.e., τ must be no more than 0.94 ns. Thus, the closed-loop amplifier must achieve a -3-dB bandwidth of at least $1/(2\pi \times 0.94\text{ ns}) \approx 170$ MHz.

If the op amp in Fig. 11.51(a) is modeled simply by a dependent current source, $G_m V_{in}$, and an output resistance, R_{out}, then the closed-loop time constant is given by (Chapter 13)

$$\tau = \frac{C_L(C_1 + C_{in}) + C_L C_2 + C_2(C_1 + C_{in})}{G_m C_2} \tag{11.21}$$

where $G_m R_{out}$ is assumed much greater than unity. This expression can be rewritten as

$$\tau = \left(\frac{C_1 + C_2 + C_{in}}{C_2}\right) \frac{C_L + \dfrac{C_2(C_{in} + C_1)}{C_2 + C_{in} + C_1}}{G_m} \tag{11.22}$$

suggesting that the op amp sees the series combination of C_2 and $C_1 + C_{in}$ in parallel with C_L, and its G_m is reduced by the feedback factor, $C_2/(C_1 + C_2 + C_{in})$ (Fig. 11.53) (Chapter 13).

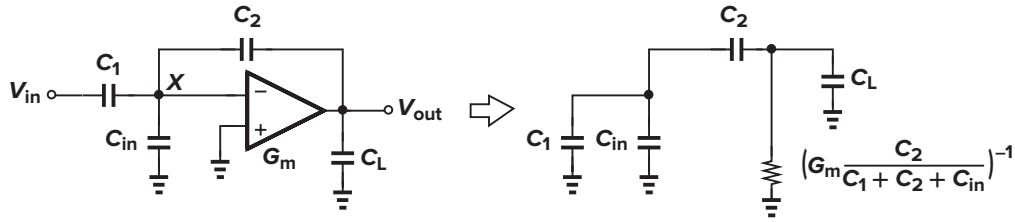


Figure 11.53 Representation of closed-loop time constant by an equivalent network.

The foregoing model is not accurate for a two-stage op amp because the internal pole (at the output of the first stage) inevitably affects the response. Let us improve our approximation by considering a frequency-compensated two-stage op amp. Recall that if the loop gain falls to 1 at the second pole, ω_{p2}, the phase margin is about 45° for unity-gain feedback.

How do we compensate the op amp for a closed-loop gain of 4 (rather than 1)? In this case, $|\beta H|$ (rather than $|H|$) must fall to 0 dB at ω_{p2} (i.e., the circuit is not compensated for unity-gain feedback). As illustrated in Fig. 11.54(a), we begin at $\omega = \omega_{p2}$ and draw a line with a slope of −20 dB/decade toward the y-axis, seeking its intercept with the plot of $|\beta H|$. We calculate the location of the compensated dominant pole, ω'_{p1}, as follows. Between ω'_{p1} and ω_{p2}, we can approximate the compensated $\beta H(s)$ as $\beta A_0/(1 + s/\omega'_{p1})$; we set its magnitude to 1 at ω_{p2}: $|\beta A_0/(1 + j\omega_{p2}/\omega'_{p1})| = 1$. It follows that

$$\omega'_{p1} \approx \sqrt{\frac{\omega_{p2}^2}{\beta^2 A_0^2 - 1}} \tag{11.23}$$

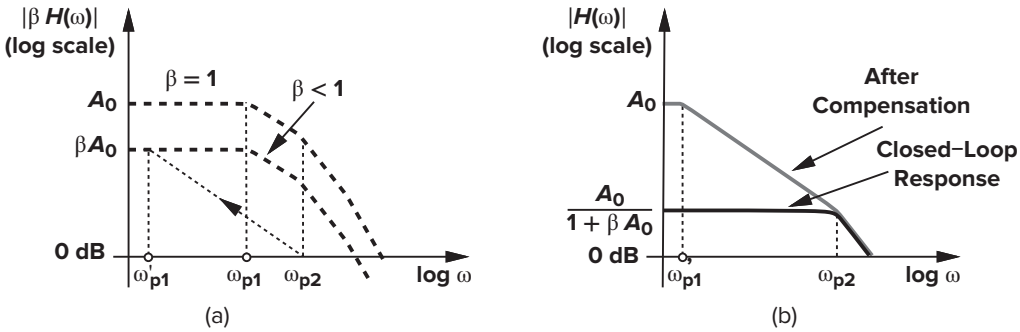


Figure 11.54 (a) Frequency compensation for loop gain of βA_0, and (b) the resulting closed-loop response.

and

$$\omega'_{p1} \approx \frac{\omega_{p2}}{\beta A_0} \tag{11.24}$$

As expected, ω'_{p1} must be chosen lower in magnitude if β increases, i.e., if the feedback becomes stronger.

▶ **Example 11.14**

We compensate an op amp for $\beta = 1/5$ and PM $= 45^\circ$. Plot the open-loop frequency response of the op amp, H.

Solution

On a logarithmic scale, the plot of H is obtained if we shift the plot of $|\beta H|$ up by $-20 \log \beta$. As shown in Fig. 11.55, H begins to fall at ω'_{p1} and reaches a value of approximately $1/\beta$ at ω_{p2}.

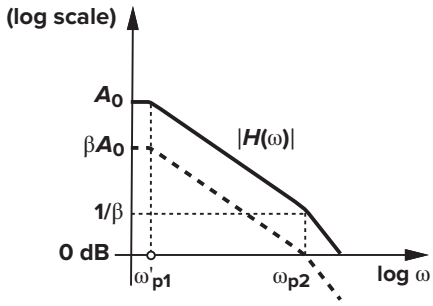


Figure 11.55

◀

Let us now construct the closed-loop frequency response with this choice of ω'_{p1}. To this end, we first plot the magnitude of the loop transmission, $|\beta H|$, for $\beta = 1$ and after compensation [Fig. 11.54(b)]. The closed-loop response begins at $A_0/(1 + \beta A_0)$ at low frequencies and begins to roll off at $\omega \approx \omega_{p2}$. From another perspective, since the ratio of the open-loop and closed-loop gains at ω'_{p1} is approximately equal to βA_0 and since the open-loop gain falls at 20 dB/dec (i.e., in proportion to ω), the two responses intersect at $\omega \approx \beta A_0 \omega'_{p1} \approx \omega_{p2}$. We therefore choose this bandwidth equal to $2\pi(170\,\text{MHz})/125 = 2\pi(1.36\,\text{MHz})$.

In summary, the closed-loop gain and settling speed requirements have translated to a dominant pole at 1.36 MHz and a second pole at 170 MHz. The open-loop gain must fall from 500 at low frequencies to 4 at the second pole. These values assume a phase margin of 45° and must be eventually revisited.

11.6.2 Op Amp Design

Based on the foregoing calculations, we seek a two-stage op amp with an open-loop gain of 500, a dominant pole at 1.36 MHz, a second pole at 170 MHz, and a differential output swing of 1 V_{pp}. We thus return to the prototype designed in Sec. 11.5.2 and see whether it can serve our purpose. Most of the specifications of that op amp are the same as those needed here. But, since the compensation can be relaxed to suit a feedback factor of 1/5, the dominant pole of the op amp need not be as low as 344 kHz. Equation (11.24) suggests that, if the feedback factor is reduced from 1 to β, then the dominant pole can increase by roughly a factor of $1/\beta$. We therefore expect that the compensation capacitor leading to the response in Fig. 11.45(a) can be lowered from 4.5 pF to 0.9 pF, raising the dominant pole frequency from 340 kHz to 1.7 MHz. For the feedforward zero to cancel the second pole, R_2 must rise by the same factor, reaching 950 Ω.

As observed in Sec. 11.5.2, the zero-pole cancellation creates a greater phase margin, allowing us to reduce C_C from 4.5 pF to 0.8 pF. However, in the present design, the feedback network capacitors also

load the output stage, lowering the *nondominant* pole, ω_{p2}. Since this effect has not been included in our calculations, we resist the temptation to reduce C_C for now and proceed to study the closed-loop behavior.

11.6.3 Closed-Loop Small-Signal Performance

Figure 11.56 shows the overall op amp and its closed-loop environment. For a nominal gain of 4, we choose $C_1 = 1$ pF and $C_2 = 0.25$ pF. With $C_{in} \approx 50$ fF, Eq. (11.17) predicts a gain error of less than 1% if $A_0 > 520$. This gain is slightly greater than that achieved by the op amp at its peak output swings (Fig. 11.41), but we deal with this issue later. For transient studies, R_F must be large enough not to cause significant "droop" during time scales of interest. Specifically, for a settling time of 5 ns, we select $R_F C_2 > 10\ \mu$s so as to confine the discharge of the capacitors to well below 1%; i.e., $R_F = 40$ MΩ. (This extremely large value implies that a switched-capacitor solution is more practical.)

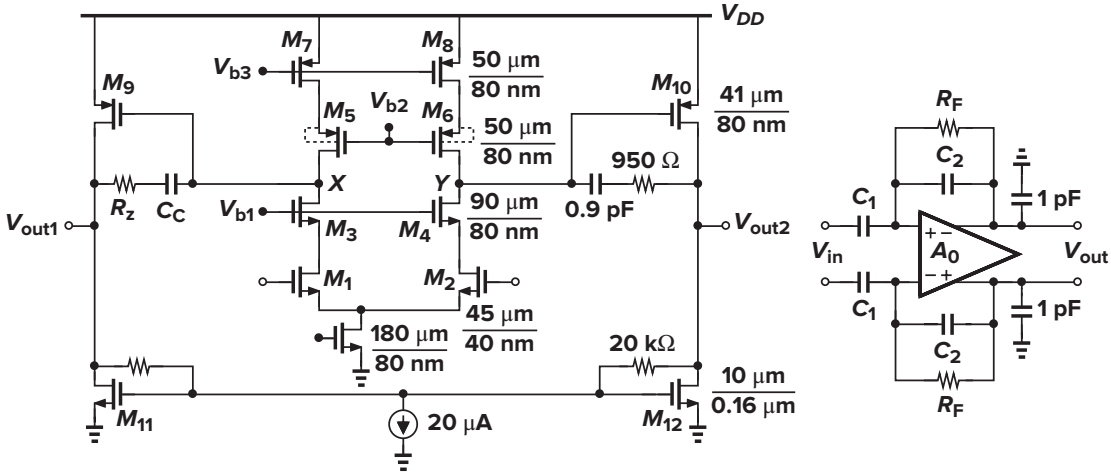


Figure 11.56 Overall compensated two-stage op amp and the closed-loop environment.

Let us apply a small step to the above circuit and examine the output behavior. With a differential input step of 25 mV, we expect an output around 99 mV (for 1% gain error). Depicted in Fig. 11.57(a) is the differential output waveform and in Fig. 11.57(b) a close-up showing the fine settling. We observe a final value equal to 98.82 mV, a result of insufficient open-loop gain.

How do we increase the gain? If we raise the length (and hence the width) of the first-stage input transistors in Fig. 11.56, C_{in} also increases, counteracting A_0 in Eq. (11.17). Instead, we double the (drawn) width and length of the NMOS cascode transistors, obtaining the output shown in Fig. 11.58. Now, the gain error is less than 1%.

▶ **Example 11.15**

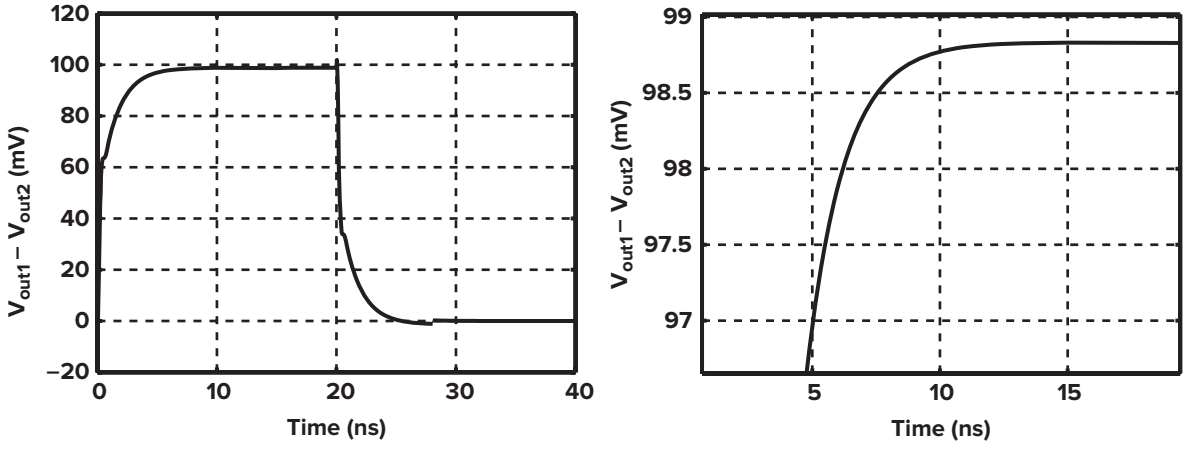
Is it possible to increase the length of the PMOS devices in the first stage to raise the gain?

Solution

Designed to provide a much higher impedance than its NMOS counterpart, the PMOS cascode structures have a weak effect on the gain of the first stage. The NMOS cascode devices, on the other hand, directly determine the voltage gain (why?).

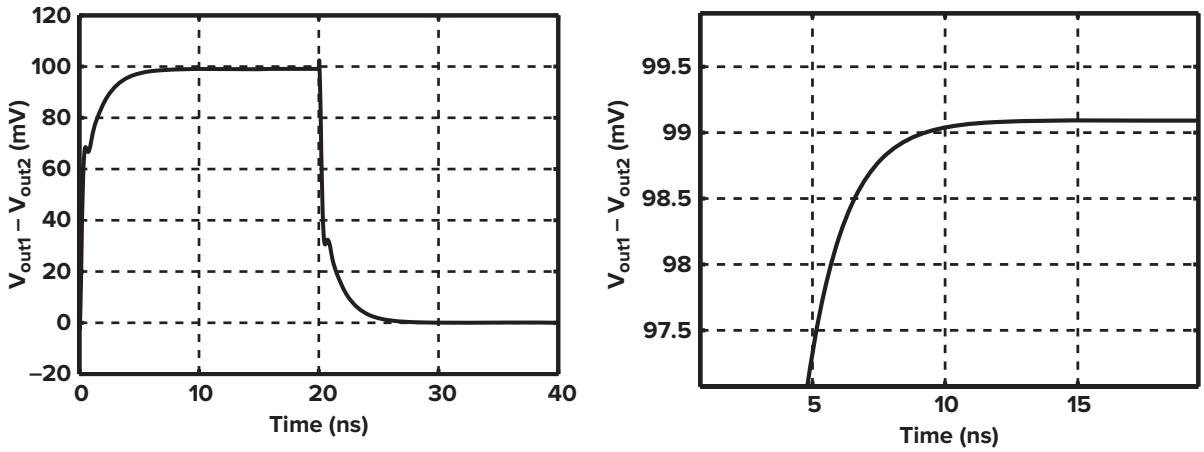
◀

Let us turn our attention to the settling behavior of the amplifier. If the output reaches 99.1 mV at $t = \infty$, how do we define the settling time to 1% precision? We must find the time at which



(a) (b)

Figure 11.57 (a) Closed-loop step response and (b) close-up showing settling to 1% accuracy.



(a) (b)

Figure 11.58 (a) Closed-loop step response and (b) close-up showing settling to 1% accuracy for $(W/L)_{3,4} = 180\ \mu$m/0.16 μm.

$V_{out} = 99.1\text{ mV} \pm 0.01 \times 99.1\text{ mV} \approx 99.1\text{ mV} \pm 1\text{ mV}$. From the waveform in Fig. 11.58(b), we obtain $t_s \approx 5.8$ ns.

In order to improve the amplifier's speed, we recognize from Fig. 11.58(a) that the circuit is "over-compensated," i.e., the output appears overdamped. We can therefore return to our choice of C_C and R_z and adjust them more aggressively. We adjust these two values, patiently explore the design space, and examine the trends in the output behavior. With $C_C = 0.3$ pF and $R_z = 700\ \Omega$, we observe the settling shown in Fig. 11.59. The settling time drops to 800 ps, a remarkable improvement. Note that R_z is *reduced* in this case, thus moving the zero to *higher* frequencies.

11.6.4 Op Amp Scaling

If the settling time is so much shorter than the required value, can we trade speed for power dissipation? As explained in Chapter 9, a straightforward approach is to perform "linear scaling." We begin with the

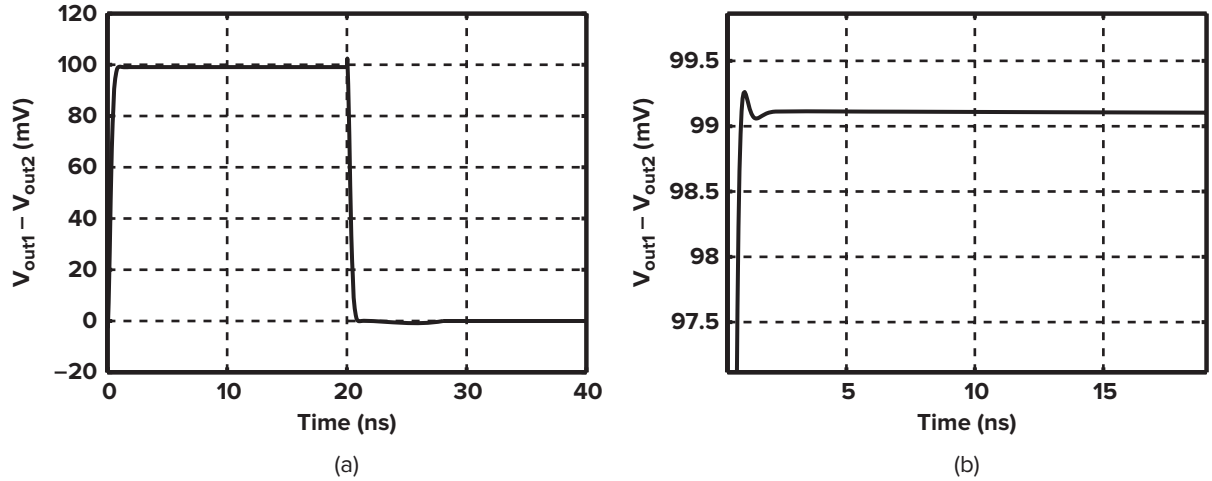


Figure 11.59 (a) Closed-loop step response and (b) close-up showing settling to 1% accuracy.

response shown in Fig. 11.60(a) and scale down all transistor widths and bias currents by a factor of α, thereby reducing the power by the same factor while retaining the voltage gain and the headroom. But how about C_C and R_z? We make four observations. (1) With the load capacitance fixed, the output pole (before compensation) is scaled down by a factor of α [Fig. 11.60(b)] because the output resistance of the second stage is scaled up by this factor (why?). (2) To maintain the same phase margin, the dominant

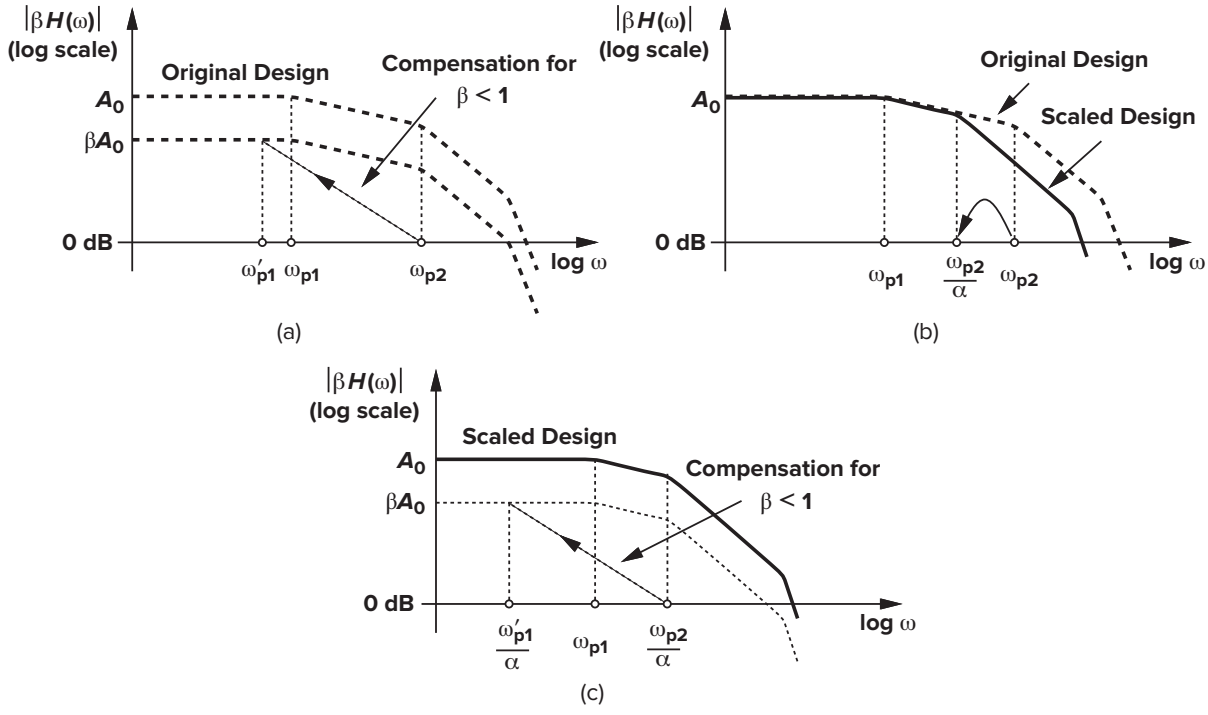


Figure 11.60 (a) Original op amp reponse and frequency compensation, (b) scaled op amp response, and (c) compensation of scaled op amp.

pole after compensation must also be scaled down by this factor [Fig. 11.60(c)]. (3) Since the output impedance of the first stage is multiplied by α, C_C should remain at its original value. (4) To place the zero introduced by R_z atop ω_{p2}/α, we multiply R_z by α (why?).

Let us try $\alpha = 2$ and examine the results. Figure 11.61(a) plots the output waveform, revealing the same final values as before and an overdamped response with $t_s \approx 2.5$ ns. We can then try scaling by another factor of 4 (i.e., $\alpha = 8$ with respect to the original design), observing the heavily overdamped transient shown in Fig. 11.61(b). Now, we adjust C_C and R_z manually to optimize the speed. With $C_C = 0.15$ pF and $R_z = 9$ kΩ, the step response appears as in Fig. 11.61(c), exhibiting $t_s \approx 4.5$ ns.

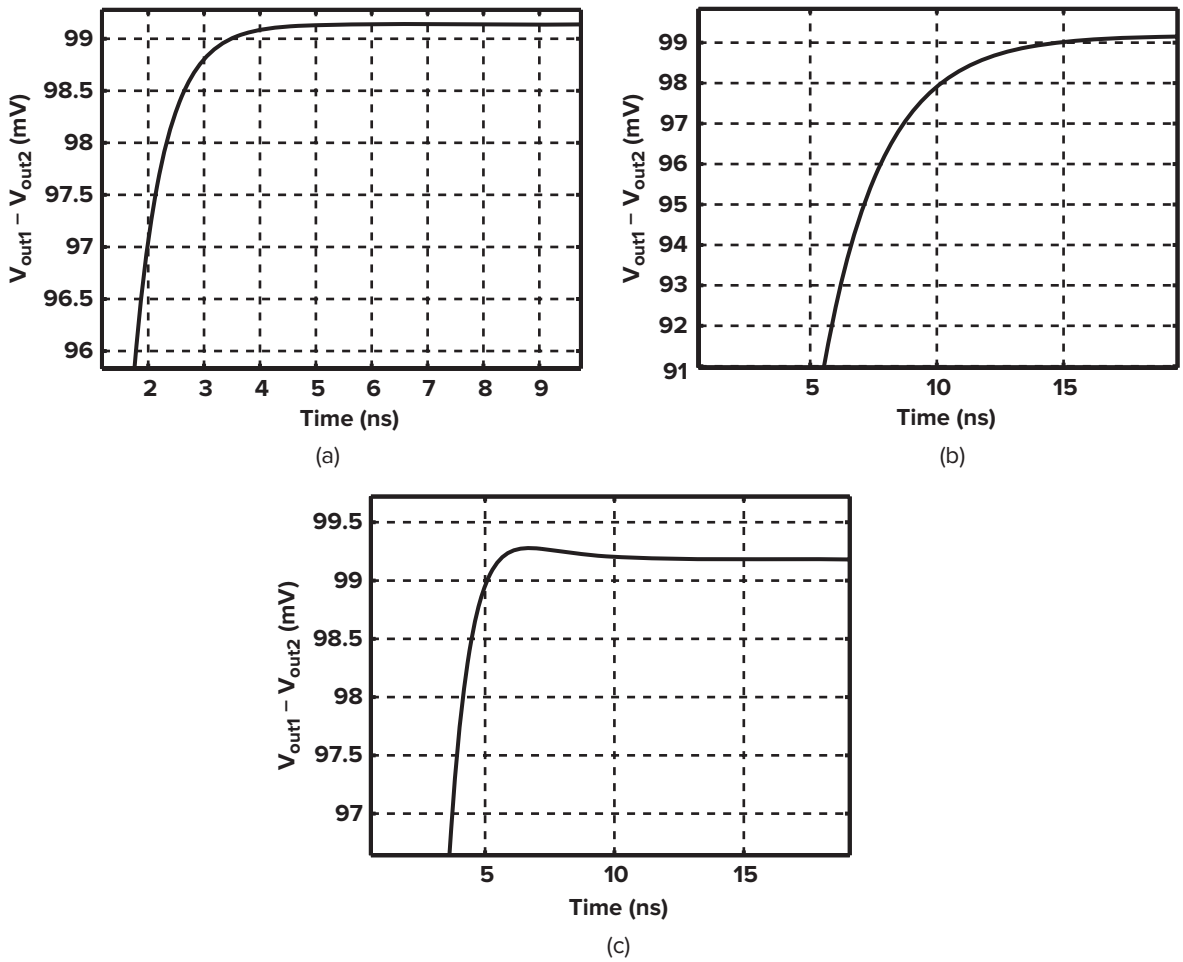


Figure 11.61 (a) Step response of op amp with scaling factor of 2, (b) step response of op amp with scaling factor of 8, and (c) modified design with $C_C = 0.15$ pF and $R_z = 9$ kΩ.

It is remarkable that linear scaling along with some adjustment of C_C and R_z affords an eightfold reduction in power (and area of the transistors). This scaling method entails minimal redesign effort because it does not alter the circuit's gain and swing values. Of course, the scaling gives rise to longer settling and higher noise (and offset). Figure 11.62 shows the scaled op amp design.[13]

[13] In practice, we do not rely on such precise values as 113 μA and 5.1 μm but round them to 115 μA and 5 μm, respectively.

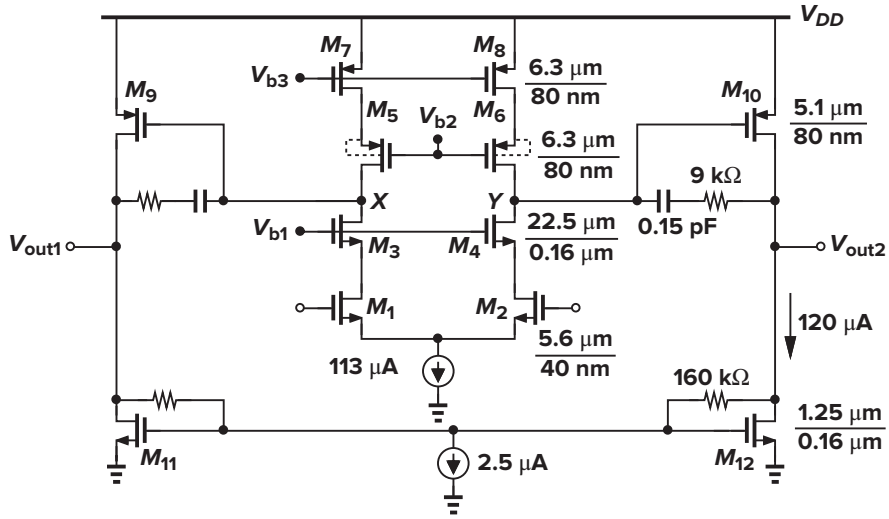


Figure 11.62 Scaled op amp design.

11.6.5 Large-Signal Behavior

The amplifier's ultimate test is with large output swings (1 $V_{pp,\text{diff}}$). Under this condition, the open-loop gain may drop as some transistors sustain less V_{DS}, and the speed may suffer as slewing may occur. In the previous simulations, the differential output begins from zero, jumps to some value, and returns to zero. For large-signal tests, however, V_{out} must swing from -0.5 V to $+0.5$ V, which can be accomplished by setting the initial differential conditions at the op amp inputs such that $V_{out} = -0.5$ V at $t = 0$. The result is shown in Fig. 11.63.

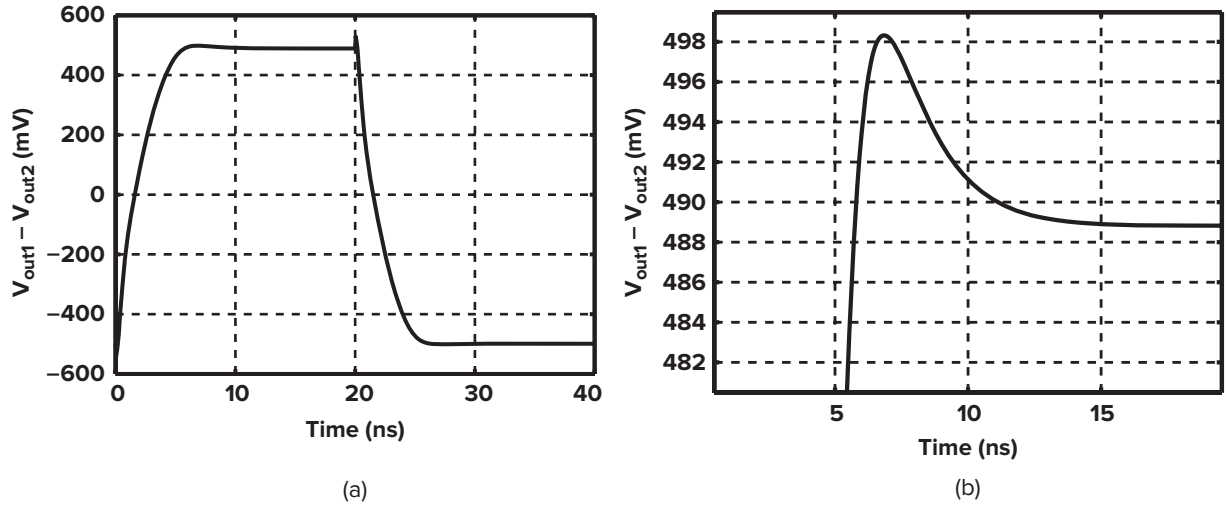


Figure 11.63 (a) Large-signal response of closed-loop amplifier, and (b) close-up of (a) showing settling to 1% accuracy.

We make two observations, First, the total change in V_{out} from $t \approx 20$ ns to $t \approx 40$ ns is equal to 987.4 mV, about 2.6 mV less than the allowed value for 1% gain error. Second, the settling to 1% from the final value is about 6 ns.

Let us first deal with the insufficient gain. We can measure the voltage gain of each stage under these conditions by dividing its differential output swing by its differential input swing (after the voltages have settled). We obtain $A_v = 39.5$ for the first stage and 10.2 for the second. (In small-signal operation, these values are equal to 46.3 and 11.2, respectively.) The open-loop gain has thus dropped from 518 to 403. To raise the gain, we double W and L of the NMOS cascode transistors ($W/L = 45\ \mu\text{m}/0.32\ \mu\text{m}$) in the first stage and the NMOS current sources in the second, arriving at the output shown in Fig. 11.64. The gain error is now less than 1%, but the settling has become longer because the pole associated with the source of the NMOS cascode transistors significantly degrades the phase margin.

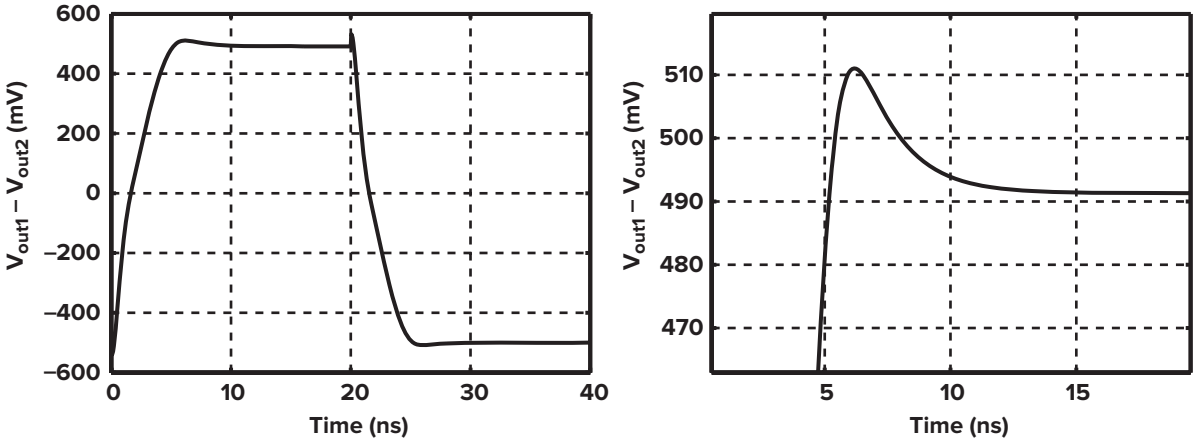


Figure 11.64 (a) Large-signal response of closed-loop amplifier, and (b) close-up of (a) showing settling to 1% accuracy.

▶ Example 11.16

Estimate the above pole frequency.

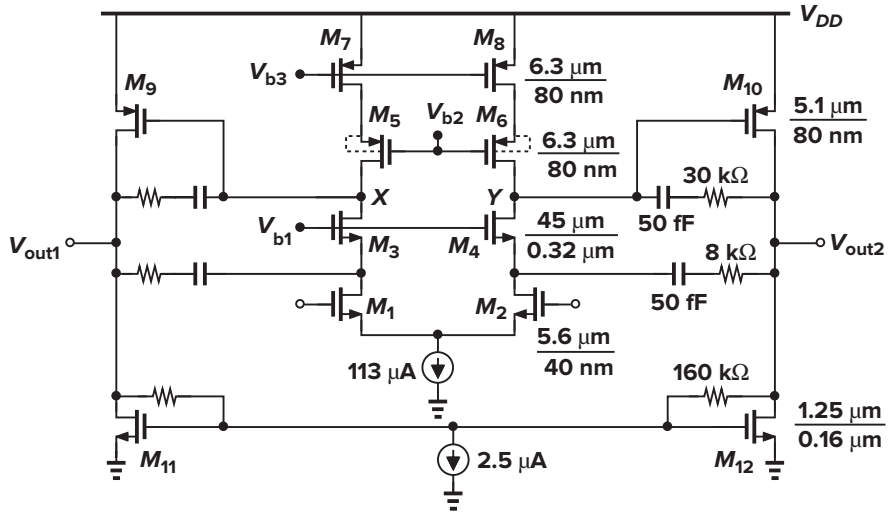
Solution

With $C_{ox} \approx 15\ \text{fF}/\mu\text{m}^2$, the gate-source capacitance of the cascode NMOS transistors amounts to $(2/3)(45\ \mu\text{m} \times 0.32\ \mu\text{m}) \times 15\ \text{fF}/\mu\text{m}^2 \approx 144$ fF. (Our calculation is sloppy because the effective length is less than 0.32 μm and the overlap capacitance is neglected.) To this value we must add the S/D junction capacitances and the gate-drain capacitance of the input transistors, obtaining roughly 200 fF. To estimate the transconductance of the cascode devices, we assume that they operate in weak inversion and write $g_m \approx I_D/(\xi V_T) \approx 56.5\ \mu\text{A}/(1.5 \times 26\ \text{mV}) = 1/(690\ \Omega)$. The pole frequency is thus around 1.15 GHz, contributing substantial phase shift at the open-loop unity-gain frequency.

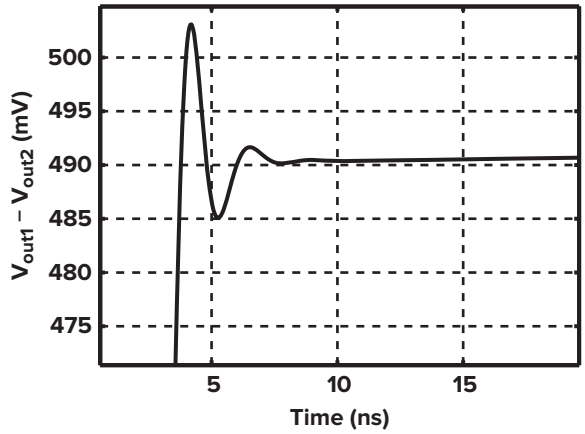
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To address the settling issue, we consider cascode compensation (Chapter 10). In fact, we can combine both methods and, with some iteration, reach the design shown in Fig. 11.65(a). Depicted in Fig. 11.65(b), takes less than 5 ns.[14] This performance is achieved with a power consumption of 370 μW.

[14] In this case, the 1% margin to the final value is equal to $490\ \text{mV} \pm 1\ \text{V}/100$ because the total swing is from -0.5 V to $+0.5$ V.



(a)



(b)

Figure 11.65 (a) Final op amp design, and (b) its closed-loop large-signal step response.

11.7 ■ Summary

This chapter has portrayed to the reader how the analog designer's mind works. We have seen that the design proceeds methodically, assuming an almost arbitrary power budget, and first strives to meet the voltage swing and gain requirements (the toughest issues today). With a reasonable design in hand, we then aggressively reduce the power by linear scaling, paying attention to parameters that cannot be scaled (e.g., the load capacitance) and bearing in mind that speed, noise, and offset degrade. Our efforts exemplify three steps in good analog design. (1) We closely examine the circuit's behavior and understand the root cause of the undesired phenomena. (2) We adjust only the circuit parameters that relate to the root cause—we do not blindly play with any random device. (3) We continue to explore various circuit techniques and new ideas, sometimes reaching a dead end but many a time improving the performance. The reader can see that we optimize the circuits "by hand" rather than automate the task using tools found in simulators. High-performance analog design requires human intelligence.

Problems

11.1. Consider the characteristics shown in Fig. 11.2. Estimate a λ value for $V_{GS} - V_{TH} = 350$ mV based on the slope from $V_{DS} = 0.2$ V to 1 V. [Hint: express the ratio of two currents at V_{DS1} and V_{DS2} as $(1 + \lambda V_{DS1})/(1 + \lambda V_{DS2})$]. Repeat this calculation for $V_{GS} - V_{TH} = 200$ mV, 250 mV, and 300 mV. What trend do you observe?

11.2. Explain why g_m falls in Fig. 11.6 as $V_{GS} - V_{TH}$ exceeds 0.5 V.

11.3. Suppose a hypothetical transistor exhibits a transconductance given by $g_m = \beta(V_{GS} - V_{TH})^2$.
(a) Find an expression for I_D as a function of $V_{GS} - V_{TH}$.
(b) Find two other expressions for g_m.

11.4. Sketch the plots in Fig. 11.7 for the device introduced in the previous problem.

11.5. We wish to bias a transistor with $L = 40$ nm at $I_D = 0.25$ mA. Referring to Fig. 11.13, determine which case yields a higher output impedance: $W = 5\ \mu$m or $W = 10\ \mu$m.

11.6. Explain what happens to the unachievable region in Fig. 11.15 if ξ falls from 1.5 to 1.0. Assume that the behavior in strong inversion does not change.

11.7. Modeling the thermal noise of M_6 in Fig. 11.21 by a voltage source in series with its gate, determine the gain that it sees as it reaches node Y. Use the exact expression for the gain of a degenerated CS stage. Compare this result with the thermal noise contributed by M_8.

11.8. Consider the arrangement shown in Fig. 11.24(b). How high can the input CM level be for M_{13} and M_{14} to remain in saturation? Does I_{D11}/I_{D12} increase or deacrease beyond this point?

11.9. Suppose a closed-loop amplifier exhibits ringing at a frequency f_1 in its step response [as in Fig. 11.46(b)]. Does this provide any information about the phase response of the open-loop circuit?

11.10. A two-stage op amp contains a nondominant pole, ω_{p2}, at the output and is compensated for PM $= 45^\circ$ so that $|\beta H|$ drops to unity at ω_{p2}. Assume that the dominant pole is much lower than ω_{p2}.
(a) Estimate the degradation in PM if the load capacitance seen at the output is doubled.
(b) How should the compensation be modified to ensure that PM $= 45^\circ$ again?

11.11. Estimate the closed-loop time constant in Fig. 11.57 and see if it agrees with the open-loop dominant frequency of 1.7 MHz.

11.12. Suppose that in Fig. 11.60, we scale an op amp *up* by a factor of α. If the load capacitance is constant, how much bandwidth improvement can be achieved?

11.13. Modeling the op amp in Fig. 11.51(a) by a voltage-dependent current source equal to $G_m V_{XY}$ and an output resistance equal to R_{out}, calculate the zero of the closed-loop transfer function. (Hint: the output voltage is equal to zero at the zero frequency.)

11.14. Consider the situation illustrated in Fig. 11.47(a). Suppose we place a capacitor, C_F, in parallel with the feedback resistor. Prove that C_F introduces a zero in the loop transmission and determine its value so as to cancel the pole created by C_{in}.